The HTTPZ Map

v16.5

What the ZAMA token could be worth in 2033 if confidential computing becomes the default on public blockchains: the evidence, the model, the arguments for and against, and what to watch

By @HuntWithBaghera · data as of 27 Oct 2026 · all dates and times UTC · not financial advice

The short version

Zama builds technology that lets public blockchains compute on data that stays encrypted the whole time. Balances, trades and payments can be private and still be checked by the network and, where the law requires, by an auditor. Apps pay a small fee in the ZAMA token every time they send encrypted data in or ask for a result to be decrypted. This report asks one question: if confidentiality becomes the normal way money moves on public chains, what does that fee flow make one ZAMA worth in 2033?

2033 ZAMA price, success case	Low (P10)	Middle (P50)	High (P90)	Chance ≥ $2,000	Value today at 20%/yr
Headline (model version 2.4, 27 Oct 2026)	$363	**$1,333**	$3,776	32.5%	$432
Same, without (e) fees on parked value	$357	$1,273	$3,608	30.5%	$412

(e) fees on parked value stated plan (not verified). **Evidence:** Zama's CEO, on the record, described "curator revenue share and stablecoin yield kickbacks" (The Rollup, 16 Sep 2026). A primary-source statement by the person who sets the fee policy is real evidence, and it fits the model's percentage-of-value fee logic. **Timing caveat:** it is not built yet. Nothing on chain contradicts it; it simply does not exist on chain today. **Verifier signal:** a confidential vault's fee recipient pointing to a Zama or burner address, or an issuer announcing a revenue share with Zama. Both readings are shown above: with (e) P50 $1,333, without it $1,273 (−4.5%).

Live fees today vs the P50 need. Organic Gateway fees run at about **$6.6k a year** (27 Oct 13:43 UTC, smoke tests excluded), with a ceiling of about $11.9k. P50 needs **$123B** in 2033, which means growth of about 15× a year for six years. That gap is the report's biggest leap of faith, argued on page 2. **What moves P50 most** (full range of each input): activity per dollar 85% · confidential by default happens 78% · free-float net selling 67% · core litepaper fee book 28% · decryption path 27% · confidential-by-default level 25% · burn share of fees 20% · tokenised value in 2033 17% · gateway share of confidential tokenised value 8% · fees on parked value 7%. **Version 2.4** (approved 27 Oct) makes four input changes, (b) to (e). Together they take P50 from $1,264 to $1,333 (+5.5%); most of that is (e), fees on parked value, a stated plan (chapter "How to read this report").

The chain behind the headline (model 2.4, 2033)

Each column shows the typical simulated future that lands at that price: the median of every draw within ±5% of it. These are the model's own intermediate values, not separate percentiles, so each column adds up. Outside reference points are at the end of chapter 3, tagged as estimates. They are context and never feed the model.

Link in the chain	Low (P10)	Middle (P50)	High (P90)

1. Tokenised assets + stablecoins on public chains	$21.3T	$28.3T	$43.4T
2. Value secured by Zama (confidential, via the Gateway)	$0.88T	$13.9T	$24.8T
Confidential share of link 1	4.3%	54%	58%
3. Gross protocol fees per year (all lanes)	$39B	$123B	$247B
Fees as % of value secured (link 3 ÷ link 2, derived)	3.9%/yr	0.88%/yr	1.00%/yr
4a. Supply locked (stake, collateral, wallets, reserve)	52%	52%	51%
4b. Supply burned by 2033 (share of gross)	0.75%	0.70%	0.52%
5. Free float that actually sells per year (share of supply) ASSUMPTION	0.43%	0.34%	0.31%
6. Supply in 2033 (after burn)	13.45B	13.34B	13.22B
7. Market cap = price × supply	$4.9T	$17.8T	$49.7T
Price	**$363**	**$1,333**	**$3,776**

How to read it. The low case is mostly the opt-in world: privacy stays a choice and only about 4% of value is confidential, so fees are about a third of the middle case. The jump from P10 to P50 is the confidential-by-default switch: confidential value goes from $0.9T to $13.9T. From P50 to P90 is mostly scale. About half of supply is locked in every column, and the free float sells only about 0.3–0.4% of supply a year into fee buyers, so fees of $123B clear at a market cap of about $17.8T (fees about 1/144 of the cap). Burn is small (under 1% of supply by 2033) because at these prices a dollar of fees buys few tokens. Supply is about **13.3B**, not today's 11.34B: staking rewards keep minting until fees cover security. Source: the litepaper sets staking rewards at "an inflation rate (5% initially), which can be changed via governance"; no multi-year schedule is published. On chain, 375.6M ZAMA has been emitted since 2 Feb 2026 (about 4.7% a year; zamadashboard tokenomics, 27 Oct). The path to 13.3B (5% until fees cover security, then 0–1.55% a year, minus burns) is the model's own estimate. **What version 2.4 changed:** four inputs, (b) to (e), net +5.5% on P50; most of it is (e), fees on parked value, which is a stated plan and not yet verified. The float-turnover input is unchanged from 2.3. **Value today, $432:** the P50 of $1,333 discounted at 20% a year over 6.2 years ($1.2^{6.18} = 3.09$). That is about 5,800× the spot price of $0.0743. It is a conditional value, not a fair price, because the model prices success, not the odds of getting there.

Fees as a share of value secured. At P50, fees are about **0.9% a year** of the value Zama secures (88 bp; 3.9% at P10, where little value is confidential but the core payment book still pays; 1.0% at P90). This ratio is derived: the model charges per operation and per trade, not per dollar held, so it is an output, not an input. Reference points, all estimate: custodian asset-servicing fees about 0.005–0.02% a year of assets; stablecoin issuers keep about 3.5–4.5% a year of their float as reserve yield; Visa and Mastercard net revenue is about 0.2–0.3% of payment volume (a flow, not a stock); Lido takes 10% of staking rewards, about 0.3% a year of staked ETH; Ethereum's fees are about 0.06% a year of ETH's market cap (DefiLlama, 27 Oct). So 0.9% sits far above custody and Ethereum, near the card networks' take, and well below what stablecoin issuers earn on float. It is high for a stock of value and only reasonable if that value moves often.

Implied price-to-fees: where 144× sits. At P50 the cap is $17.8T on $123B of fees, about **144×**. Live multiples today (market cap from CoinGecko ÷ last 12 months' fees from DefiLlama, pulled 27 Oct 15:07 UTC; holder revenue = burn or buyback in brackets): Ethereum 1,617× (5,846×), BNB 590× (5,897×), Solana 250× (2,228×), Chainlink 140× (149×), Tron 92× (92×), Hyperliquid 20× (26×), Jupiter 5× (27×), Aave 3× (99×), Lido 1× on gross staking rewards (44×). In plain words: **144× is in the middle of the range, not above it.** It is far richer than the fee-and-buyback tokens (Hyperliquid, Jupiter, Aave) and about level with Chainlink and Tron, but far below the base-layer chains, which trade at 250–1,600× fees. ZAMA today trades at about 6,600× its full fee meter and about 29,000× organic fees ($190M cap). One honest difference: peers use circulating cap, the model uses all 13.3B tokens. **Mechanism:** 144× is not a P/E the model assumes. Price clears where fee buyers meet sellers. About half of supply is locked (stake, collateral, balances), and the free float is assumed to net-sell only about 0.7% of the float a year, which is 0.34% of total supply ASSUMPTION: tau, free-float net selling, 0.5–1.5% of the float a year (about 0.25–0.75% of total supply). With so few tokens for sale, $123B

of fee demand clears at a high price. If the float sold more, the multiple would fall toward the buyback tokens; this input moves P50 by 67%.

Activity per dollar, the top driver. Today: about 189 organic input checks a day after removing smoke tests (623) and Merkl reward claims (≈23% of the rest), on about $227M of confidential non-ZAMA value (zamadashboard, 27 Oct 15:08 UTC; vault shares counted near $1). That is about **0.0003 operations per dollar a year**. If every one of the 1,932 daily decryptions were organic, it would be 0.0034. The model's 2033 P50 implies fees of 0.88% of value at $0.0145–0.025 an operation, about **0.35–0.6 operations per dollar a year** (its own activity input is a 1–3× multiplier on the litepaper fee book). So the model needs each confidential dollar to be about 100–2,000× more active than today, as well as about 60,000× more value. Plain path: today's value is parked yield vaults that barely move; the model needs it to become payments, settlement and collateral that moves weekly. What we could not remove: decryptions are not split between smoke tests and users, one placeholder user (17 inputs) is kept in, and vault value is itself lifted by Merkl incentives ($43M in one campaign).

From today's live fees to 2033: the gap, owned

Fee meter (Gateway, per year)	Value	Source
Today, organic inputs (excludes Zama's own smoke tests)	**≈$6.6k**	245 paid input checks in 24h from 80 non-Zama users, 27 Oct 13:43 UTC. Without one unresolved placeholder user: ≈$6.2k
Today, ceiling (if every decryption is organic)	≈$11.9k	plus 1,932 decryptions (193 ZAMA a day), not yet split between smoke tests and organic use
Today, headline including smoke tests	≈$28.8k	2,800 paid ops a day. Zama's smoke tests are 623 of 868 inputs (72%) and are not counted as demand
Recent trend, organic inputs	$2.5k → $6.6k	24 Oct to 27 Oct (24h windows, noisy)
2033, model P50	$123B	chain above

The gap. $6.6k to $123B is about 19 million times, which means fees growing about 15× a year (14× from the $11.9k ceiling) for six years. No protocol has done that. It is the single largest leap of faith in this report. **The case for it:** today's meter counts a product that has not been switched on. Fees are set to start free and be captured later, per-op prices are the litepaper's, and the volume that counts is institutional settlement, which is not on public chains yet. **The case against:** about a quarter of today's organic inputs are claims of Zama-funded rewards (63 of 270 on 27 Oct). One campaign's daily incentive (≈$11.7k) is about 690× the daily organic input fee. Usage may fall when incentives end. **What we watch:** organic inputs per user after the 3, 8 and 12 November campaign ends (12 Nov 23:59 UTC is the end of Merkl's Flowdesk USDT campaign, 1.36M ZAMA, per Merkl's public API (data interface), 27 Oct). Signals about real use are cUSDT (8,830 holders) and cUSDC (2,006). Stablecoin wrappers and the vault router are 114 of 245 organic inputs (47%). Since version 2.4 the model's ramp starts from a live organic band of $6.6k–$11.9k: the 27 Oct reading up to the ceiling. On its own the change moves P50 by +0.3%, because it only shifts when the hand-off happens.

What moves the middle number most

Each input is pushed to the ends of its own range while everything else stays the same (12,000 draws per world, base P50 $1,309 at that draw count). The bar shows the total swing.

Input	Low end → high end	Swing in P50
Activity per dollar (composability)	−30.1% → +55.0%	 85%

Confidential by default happens (odds 0 vs 1)	−54.0% → +24.4%	78%
Free-float net selling (1.5% vs 0.5% of the float a year)	−28.5% → +38.6%	67%
Core litepaper fee book	−12.3% → +15.5%	28%
Decryption path (all off-Gateway, unbilled vs all via Gateway)	−18.8% → +8.7%	27%
Confidential-by-default level (60% vs 95%)	−16.5% → +9.0%	25%
Burn share of fees (50% vs 100%)	−9.6% → +9.9%	20%
Tokenised value in 2033	−6.9% → +10.4%	17%
Gateway share of confidential tokenised value	−3.8% → +4.1%	8%
Fees on parked value, (e) ($2B vs $11.6B) stated plan	−3.3% → +3.6%	7%
Fees on parked value (stated plan)	−3.3% → +3.6%	7%
Staker yield requirement	+3.9% → −1.9%	6%
Security budget (hand-off timing)	+1.4% → −1.3%	3%
Price-fee elasticity	−0.3% → +1.0%	1%
TEE (trusted execution environment: secure hardware) version of the ERC-7984 confidential token standard (cut 3% vs 10%)	+0.6% → −0.5%	1%
Stablecoin supply in 2033	−0.2% → +0.3%	0%
Operator stake share (5% vs 6.5%)	−0.1% → +0.1%	0%
Per-op fee drift (hardware cost)	−0.0% → +0.2%	0%
Collateral ratio (50 vs 10)	−0.0% → +0.0%	0%

Adoption drives it, not the token mechanics. The largest inputs are activity per dollar and whether confidential by default happens at all, the switch between the P10 world and the P50 world. Next is how much of the free float net-sells into fee buyers each year (0.5–1.5% of the float, about 0.25–0.75% of total supply), unchanged since 2.3. The four 2.4 inputs are all small: fees on parked value move P50 by about 7% across their range, the TEE cut and operator stake by under 2%. Stablecoin supply, collateral ratio and per-op fee drift barely move it. The vision gets sharper where evidence shows up on composability and on the confidential-by-default switch (chapter 17).

The February 2027 unlock does not move this number. The headline is a fee-driven value for 2033. The unlock, and whether it is released daily or monthly, affects the price path and timing over the coming months, not the end-state value (chapter 14).

P10, P50 and P90 mean: in 10%, 50% and 90% of the simulated futures the price comes out at or below that number. P50 is the middle outcome. Spot price at the time of the run was $0.0706. The model is conditional: it assumes crypto and confidential finance succeed, and asks what success is worth. It does not price the chance of failure.

What is new in this edition

- **v16.4–16.5:** the free-float selling rate now uses one unit everywhere (share of the float a year); "default-dark" renamed "confidential by default" so it cannot be confused with the "dark period" outage scenario; sensitivity figures rerun and matched across table and text; acronyms spelled out at first use.
- **v16.3:** model version 2.4 with four input changes, (b) to (e): P50 moves from $1,264 to $1,333 (+5.5%). A proposed float-turnover change was reverted before publication and is kept only as a labelled what-if, (a′), on the stake side. A passage on fees on parked value versus fees on flows. Corrected reading of the Foundation's monthly staking batches. Every table and chart has been rerun. The sources in the encrypted shared state section have been re-checked.
- **v16.2:** the chain behind P10, P50 and P90 on page 1, with outside reference points; live fees and the growth needed to reach 2033; a ranking of what moves the middle number; upside beyond finance (not in the headline); corrected Arcium, Inco and Circle evidence; and new on-chain signals, each tied to one input. No model input changed in v16.2.
- **A sharper argument for why the world needs this.** We now test every weak early signal against two questions: is there a real need to compute on shared encrypted data, and is anyone other than FHE (fully homomorphic encryption: computing directly on encrypted data) meeting that need? Where the need holds and no rival is winning, slow early data is read as a matter of timing, not as evidence against.
- **A rival-by-rival audit** of the comparison table in Zama's own litepaper, with sources, plus a map of which technology wins each job today and where each ends up.
- **New evidence:** confidential tokens live in the Zerion wallet; S&P Global's acquisition of OpenZeppelin, co-author of the confidential token standard; a stated compute cost far below today's fee (about 200 times below the litepaper input price); a full read of Zama's 246 blog posts.
- **A rerun of the exploit-wave path test** (a 6–9 month wave of AI-driven hacks), and four new what-ifs, all kept beside the headline.
- **Corrections**, including one to our own earlier reading of the February 2027 token unlock.

The three biggest open questions: (1) how decryption will be priced as it moves to plain web requests; (2) how much confidential value stays in Zama's vaults when incentives end on 3, 8 and 12 November; (3) when issuers on the T-REX tokenisation network actually switch privacy on.

How to read this report

This chapter explains the labels and rules used throughout, so a reader can tell a fact from a plan from an argument.

Evidence labels. Every claim that matters carries one of three tags:

- verified seen on chain, in published code, or in a primary document.
- stated plan something Zama or a partner has said it will do and is visibly working on. We write these with confidence, because a success map should take a company's stated, in-motion plans as its base. But they are not facts yet, and the tag keeps them separate.
- inference our own reading of the evidence.

Dating rule. Newer and more detailed statements beat older launch wording. Code and live use beat talk. Work on an unmerged code branch counts as direction, not final design.

What-ifs sit beside the headline, never inside it. The headline only contains Zama's published fee schedule, stated plans and our sourced estimates. Everything else is written as an argument with a claim, a causal chain, evidence, signposts to watch and the strongest counter-argument.

The model only moves on evidence. Version 2.3 counted each piece of evidence once, in the one input its mechanism acts on (box in chapter 4). Version 2.4 (27 Oct 2026) makes four changes, (b) to (e). Each feeds one input. Change (a), a proposed rereading of float turnover, was withdrawn before publication (see below). Step by step, applied in order:

Step	P10	P50	P90	≥ $2k	P50 step
Version 2.3	$351	$1,264	$3,625	30.3%	—
(b) Live organic fee base $6.6k–$11.9k (was $5.2k–$10.6k)	$352	$1,268	$3,636	30.5%	+0.3%
(c) Operator-vault stake bonded apart from free stake (5–6.5% of supply)	$358	$1,282	$3,646	30.8%	+1.2%
(d) TEE version of ERC-7984 takes 3–10% of Zama's Gateway share	$357	$1,273	$3,608	30.5%	−0.7%
(e) Fees on parked value, $2–11.6B a year at P50 scale stated plan = **2.4**	$363	**$1,333**	$3,776	32.5%	+4.7%

Each change alone, from 2.3: (b) $1,268 (+0.3%); (c) $1,279 (+1.2%); (d) $1,255 (−0.8%); (e) $1,330 (+5.2%).
Why (a) was withdrawn. A reviewed proposal read "net absorption of 5–30% a year" as tokens the free float sells into fee buyers. Its proposer confirmed it means the opposite: holders net-adding tokens to stake. That shrinks the float rather than adding sellers, so float turnover stays at its 2.3 value. The stake-side reading is run as what-if (a′) in chapter 13: 5–15% a year of the free float moving into stake, counted on free stake only so operator stake (c) is not counted twice. It adds +1.4% and stays out of the headline, because on-chain data so far shows outside holders adding at most about 15% a year, lumpily (chapter 14).

> **Fees on parked value versus fees on flows.** Zama's CEO has described "curator revenue share and stablecoin yield kickbacks" (The Rollup, 16 Sep 2026) stated plan, not verified: a cut of the yield earned on value that sits confidential, not only a fee each time it moves.

> **Thesis: parked value is the bigger base.** At P50 about $14T sits confidential through Zama. Even a small cut of the yield on a stock that size is billions a year, and it is paid whether or not the value moves.
>
> **Antithesis: Zama gets a share of a share.** Only part of parked value earns yield (we use 30%), the yield is a few percent (3.5%) and Zama's cut is a fraction of that (7.5%). 30% × 3.5% × 7.5% is about 0.08% of value a year: $11.6B at the centre, about a quarter of the $46.3B per-op flow meter. For parked value to out-earn flows, the cut would need to exceed about 30% of the yield, more than curators or issuers have any reason to give away.
>
> **Synthesis: an add-on, not a second engine.** Fees on parked value enter the model as $2–11.6B a year at P50 scale, scaled per draw by confidential value, in the core fee line only (+5.2% alone). They scale with the same confidential value as the flow fees, so they strengthen the confidential-by-default bet rather than diversify it. A $26B case is a what-if only. It becomes verified when a confidential Morpho vault's fee recipient points to a Zama or burn address, or an issuer announces a revenue share. Today those vault fees are zero.

1. Method: the two-question test, and the case for FHE

This chapter explains the technology in plain words and sets out the test we use to read ambiguous early data. It matters because most of the debate about Zama is really a debate about whether early slowness means "not yet" or "not ever".

Why this map steelmans. This report deliberately argues the best plausible case. A new, close to revolutionary technology has no history to average over, so a "typical outcome" would be a guess dressed up as caution. The downside is not hidden. It sits beside the case in P10, in the what-ifs (chapter 13) and in a steelmanned counter-case in every chapter. This is a stated method, not a bias. At this stage most of the evidence is stated plans from primary sources: the team and its partners. That is normal for a protocol this new. The map uses those plans as evidence, tags them, and names the measured signal that will harden or correct each one in later versions.

The technology in one page

Fully homomorphic encryption (FHE) is encryption that lets a computer add, compare and multiply numbers without ever decrypting them. The result comes out encrypted, and only someone holding the right key can read it. The idea was an open problem for thirty years until Craig Gentry showed it was possible in 2009 ("Fully homomorphic encryption using ideal lattices", STOC 2009). For a decade it was far too slow for real use. Zama's contribution is engineering: a fast FHE scheme (TFHE) and tools that let ordinary smart-contract developers use it.

FHE schemes are built on **lattices**: mathematical problems about finding short vectors in huge grids of points (formalised as "learning with errors" by Regev, 2005). No

known algorithm, classical or quantum, solves them efficiently. That is why the US standards body NIST chose lattice schemes for its first post-quantum encryption standard (FIPS 203, ML-KEM, August 2024). **Post-quantum** means the encryption is expected to survive a future quantum computer. That matters for financial records, which must stay secret for decades: data copied today could be decrypted later if it is protected by older math.

The rivals, briefly:

- **Zero-knowledge proofs (ZK)** let a user prove something is true (for example "I have enough balance") without revealing the data. Excellent for one person's own data. Hard when many parties' hidden data must be combined.
- **Trusted execution environments (TEE)** are sealed areas inside a chip (Intel SGX/TDX, AMD SEV-SNP) where code runs privately. Fast, but you must trust the chip maker and the physical machine.
- **Multi-party computation (MPC)** splits a secret among several computers so no single one sees it. It works on shared data, but relies on the parties not colluding and is hard for outsiders to verify.
- **Private or permissioned ledgers** (for example Canton) keep data private by only sharing it with the parties to a trade. Not open or public.

Zama's network has a few moving parts that appear throughout. **Coprocessors** are off-chain computers that do the heavy encrypted math for a host blockchain such as Ethereum. The **key management service (KMS)** holds the decryption key, split among 13 independent operators: at least 9 must take part to decrypt anything, and the key stays safe unless 5 or more collude (a "threshold" key). The **Gateway** is Zama's coordination chain, where encrypted inputs are checked and decryptions are requested, and where fees are paid. **ERC-7984** is the open token standard for confidential tokens, co-written with OpenZeppelin, the most widely used library of smart-contract code. **ERC-3643**, also called **T-REX**, is the leading standard for compliant tokenised securities.

The two-question test

Early adoption data is thin and ambiguous. To read it without wishful thinking in either direction we ask two questions:

1. **Does the world need to compute on encrypted shared state?** Not just hiding one balance, but lending checks on hidden collateral, encrypted order books, sealed-bid auctions, compliance rules on hidden balances.
2. **Does FHE uniquely provide it?**

If the need holds and no rival is winning the same job, weak early data is a **timing** signal: it moves dates, not the thesis. It counts **against** only when a rival is shown to win the same job. The rule must cut both ways, so we write down in advance what would count against (see chapter 12).

Question 1: the need. Our answer is yes verified sources:

- **Funds:** visible positions invite front-running and copying (BIS Bulletin No. 58).
- **Banks:** client confidentiality is a legal duty (Swiss Banking Act art. 47; Luxembourg financial sector law art. 41; GDPR art. 5 and 17). BIS research notes that pseudonymous addresses "cannot fully ensure privacy".
- **Markets:** EU market-abuse and MiFID rules treat order and position data as confidential.
- **Companies:** salaries and treasury moves; confidential payroll has run live on Zama (via Bron) since January 2026.
- **Institutions in their own words:** J.P. Morgan's Kinexys (2024): "enhanced privacy measures are essential"; BlackRock's Robert Mitchnick on keeping KYC/AML (know-your-customer and anti-money-laundering) checks while needing "to be able to preserve privacy"; Apex and T-REX calling privacy "core infrastructure" (24 Mar 2026).

The honest limit: need is not the same as paying now. J.P. Morgan and MIT write that front-running and sanctions exposure "cannot be fully addressed through technical means alone", and BIS calls these safeguards "untested under stress". The need is partly legal and organisational. That again argues for timing, not for failure.

Question 2: uniqueness. Taken across every job, FHE is not the only option. For the hardest job, public, composable shared state that regulators can look into, with no hardware trust and post-quantum protection, it is the strongest fit, and the rivals' gaps there are built into their design. Chapter 2 makes that argument in full.

Why finance needs encrypted shared state argument, does not feed the headline

Thesis: finance runs on shared state that no one party owns. Settlement, collateral and composability all need a record that every party can rely on and act on at once. Settlement must be final for both sides. Collateral pledged to one lender must be visibly unavailable to another. A position used in one product must be usable in the next without asking anyone's permission. The BIS calls this a "unified ledger": a shared, programmable venue where money and assets sit together, so that sequences of transactions settle simultaneously and can be bundled ("composability") (BIS Annual Economic Report 2023, ch. III, 25 Jun 2023). Note that the BIS anchors its ledger in central-bank money and trusted operators, not in a public chain. That is the antithesis below, in the BIS's own words. Bears on: the tokenised value and activity-per-dollar inputs. Activity per dollar is the largest swing in the model (85%).

The problem: public shared state exposes everything. On today's public chains every balance, position and pending order is visible. Bots read pending trades and trade ahead of them. Daian and co-authors documented this in 2019: bots paying high fees to frontrun ordinary users' trades ("Flash Boys 2.0", arXiv April 2019; IEEE Symposium on Security and Privacy 2020). No bank, fund or treasurer can put a book or order flow there. That is why the T-REX network ships confidentiality as an opt-in

feature still in development. Bears on: the confidential-by-default odds, the second-largest swing (78%). The question that input asks is whether this exposure gets fixed on public chains or moves the business off them.

Why FHE specifically. The fix has to keep state shared and computable while it stays secret. FHE does both: anyone can add to an encrypted balance or match an encrypted bid, and only authorised keys (Zama's 13-party KMS, 9 needed to decrypt) can read the result (Gentry, "Fully homomorphic encryption using ideal lattices", STOC 2009; Zama protocol docs). ZK proves facts about private state that one party holds, but it does not let many parties compute on the same hidden state. That is why ZK wins one-party transfers (chapter 2). TEEs do share and compute, but they ask everyone to trust the chip maker and the enclave. TEE.fail (Georgia Tech and Purdue, with Intel and AMD bulletins of 28 Oct 2025) extracted keys from Intel TDX and AMD SEV-SNP with a memory-bus probe that costs under $1,000 to build. It needs physical access to the server, and its authors acknowledge gifts from Zama among others. MPC shares state across a fixed committee, at a cost in latency. Bears on: the Gateway's share of confidential value and the decryption path, because every encrypted input and decryption is a billable Gateway op.

Antithesis, steelmanned: permissioned ledgers are enough. Banks already get shared, private state from ledgers they run among known parties: Canton, and Kinexys at JPMorgan. Privacy comes from the access list, not from cryptography, and is cheaper and faster today. Regulators know whom to call. Most tokenised value so far sits on exactly such systems, and Circle plans privacy on its Arc chain with a TEE rather than FHE. If that holds, confidential value never reaches public chains in size, and the model's P10 opt-in world is the right one.

Synthesis. Permissioned ledgers solve privacy by shrinking who shares the state, which gives up the reason for a unified ledger. Collateral and positions cannot compose with assets outside the club without a bridge back to a trusted operator. They win where one consortium owns the whole flow. Encrypted public state wins where value has to move across institutions that do not trust a common operator, such as stablecoins, cross-issuer collateral and open markets. How much of 2033 value is of that second kind is exactly what the confidential-by-default odds and the Gateway share encode. The argument says why those inputs are not zero. It does not set their size. The headline keeps the ranges in chapter 3 unchanged.

2. FHE against its rivals: today's leads and the end state

This chapter audits the comparison table in Zama's litepaper cell by cell and then asks where each technology ends up, not just where it stands today. It matters because the price of ZAMA in 2033 depends on the end state, not the 2026 scoreboard.

Does Zama's own table hold up?

Zama's litepaper compares FHE with other FHE stacks, MPC, ZK, TEE and private chains on six rows: secure, decentralised, verifiable, composable, scalable and easy to use. We checked every cell against outside evidence. In summary:

Row	Verdict on Zama's own tick	Notes
Composable	**Fair, and understated** verified	Shared encrypted state across contracts on Ethereum, live. This is the real moat and deserves more weight than one tick.
Secure	Fair against outsiders	Lattice-based, post-quantum for the data. Per Zama's docs, decryption needs at least 9 of the 13 operators to take part, and the key stays safe as long as no more than 4 collude (5 colluding could reconstruct it) verified: Zama KMS docs: a threshold-trust model, not trustless.
Scalable	stated plan	1,040 transfers per second on GPUs, announced July 2026, a year ahead of plan; Zama projects 10,000/s by 2028 and 100,000/s by 2030 if FHE keeps getting about 10× faster every 18–24 months. The 1,040 figure measures confidential computation on one GPU node, not end-to-end settlement.
Decentralised	stated plan	Today 18 named, permissioned operators across the network's roles (13 of them run the key-management service that holds shares of the decryption key). Open, permissionless operators are the stated plan.
Verifiable	stated plan	Today it rests on a majority of coprocessors and on AWS Nitro hardware enclaves in the key system, which the litepaper itself calls "suboptimal". Cryptographic verification (ZK-MPC) is the stated fix.
Easy to use	Fair relative to rivals	Ordinary Solidity; a wallet like Zerion integrated with little code.

Two things the table leaves out work in Zama's favour: **post-quantum protection of the data** (lattice-based and expected to survive quantum computers; some ZK designs, such as hash-based STARK proofs, and MPC built on secret sharing can be post-quantum too, but TEEs cannot, and only FHE combines it with shared encrypted state) and **programmable compliance** (view keys that let a named auditor or regulator see specific data). Two things it understates work against it: **Canton's lead** in live institutional volume today, and **ZK's maturity** for simple transfers. The table is also generous to some rivals, for example ticking MPC as "scalable" while Zama's own text says MPC does not scale well for heavy compute.

The TEE question, verified

The case against hardware enclaves is now well documented. In October 2025 researchers from Georgia Tech and Purdue published **TEE.fail**: with a memory-bus probe built for under $1,000 from second-hand parts, they attacked DDR5 server memory (earlier attacks, WireTap and BatteringRAM, needed older DDR4) and extracted secret keys, in some cases attestation keys, from Intel SGX and TDX and from AMD SEV-SNP on fully updated server machines, and showed how forged attestation could be abused against real crypto deployments verified: tee.fail; Intel announcement INTEL-2025-10-28-001; AMD bulletin AMD-SB-3040. Intel said physical attacks remain out of scope; AMD said it does not plan mitigations. It follows a long series of published attacks on SGX since 2018. Disclosure: the TEE.fail page acknowledges research gifts from Qualcomm and Zama, so a Zama rival-critique sits on research Zama helped fund; the results are peer-reviewed and reproducible either way.

The fair counterpoint: these are physical attacks, so a TEE run by a known operator in a guarded data centre is much harder to hit. And Zama's own key system uses AWS Nitro enclaves as a backup layer, under the 9-of-13 threshold. The difference is that in Zama's design breaking one enclave does not reveal the data; in a TEE-only design it can.

Thesis, antithesis, synthesis

> **Thesis.** FHE becomes the default for confidentiality on public chains. Its weaknesses, speed and cost, are engineering problems that hardware solves: GPUs today, dedicated FHE chips on the roadmap, and Zama's own figure puts compute cost per confidential transfer under $0.00005. The rivals' weaknesses are structural. ZK cannot hold state that many users share and compute on. A TEE must trust the chip maker and the physical box. MPC must trust that its node set does not collude and is hard for outsiders to verify. Permissioned ledgers give up openness.
>
> **Antithesis, steelmanned.** Markets rarely pick the strongest technology; they pick the one that is good enough, cheap and here now. ZK already wins simple private transfers on Solana, natively and free. Arcium (MPC) launched its token on Solana on 22 June 2026. On 9 June Arcium itself said it had passed 1M computations and was running more than 200,000 a day. Its own explorer now shows about 2.70M in total, which averages about 12,000 a day since then, so 200,000 was a peak, not a steady rate. Circle plans opt-in privacy on its Arc chain using a TEE (AWS Nitro Enclaves at first). It is still at testnet stage, not live. Inco now describes itself as a private rollup built on TEEs, and its confidential tokens went live on Base on 2 Sep 2026 inference: Inco began on FHE. iExec runs the ERC-7984 standard on TEE, so the standard can spread without Zama earning the fees. Canton dominates bank-internal tokenisation. Network effects favour whoever gets there first, and "hardware is coming" has been said about FHE for a decade.

> **Synthesis.** Both are true on different clocks. Today's leads are real, but they sit in jobs that do not need shared encrypted state (simple transfers), in closed settings (bank books), or in places where FHE is not yet fast enough (high-frequency trading). The test that matters is: does a rival still win a job once FHE is fast and cheap? Only that counts against FHE. On that test, we read today's rival leads as timing or niches, and expect FHE to be the default on public chains across most jobs. We keep the antithesis alive through pre-registered signposts (chapter 12): Arc privacy carrying real USDC volume, Arcium usage holding up after Zama arrives on Solana, or a standards body adding a non-FHE privacy route.

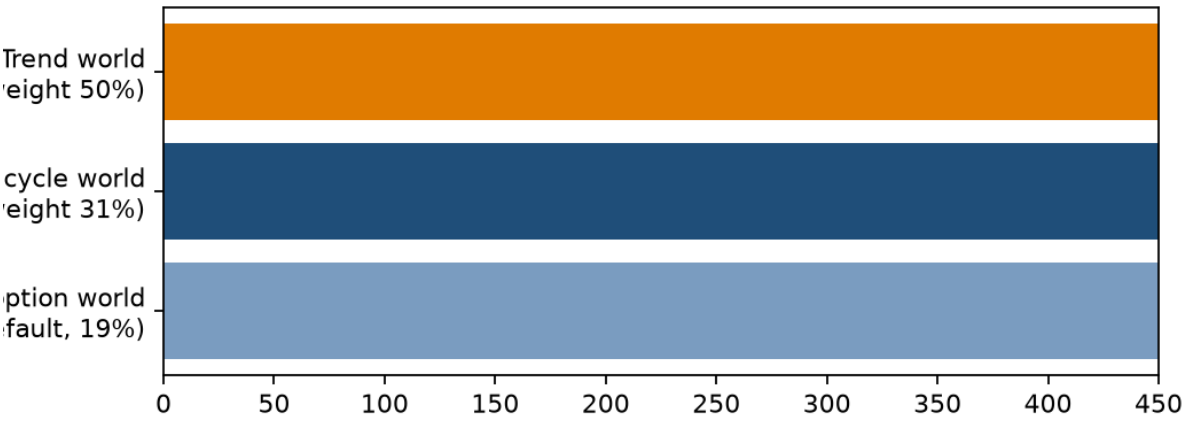
Jobs map: who wins each job

Job	Leader today	End state (once FHE is fast and cheap)
Simple private transfers	ZK (Solana confidential balances, ZK rollups)	FHE where transfers need to interact with shared state; ZK remains strong for pure one-to-one transfers
Shared-state DeFi (lending, vaults, order books)	Zama on EVM (Ethereum-compatible) chains; Arcium on Solana	FHE
Tokenised real-world assets	Canton (permissioned)	FHE on public chains; Canton for bank-internal books
Payroll and corporate treasury	Small (Bron on Zama)	FHE
Private voting	ZK	FHE (tallies computed on encrypted votes)
Fast order books, AI workloads	TEE	FHE once dedicated chips arrive, starting with sealed auctions

3. The success map and the headline

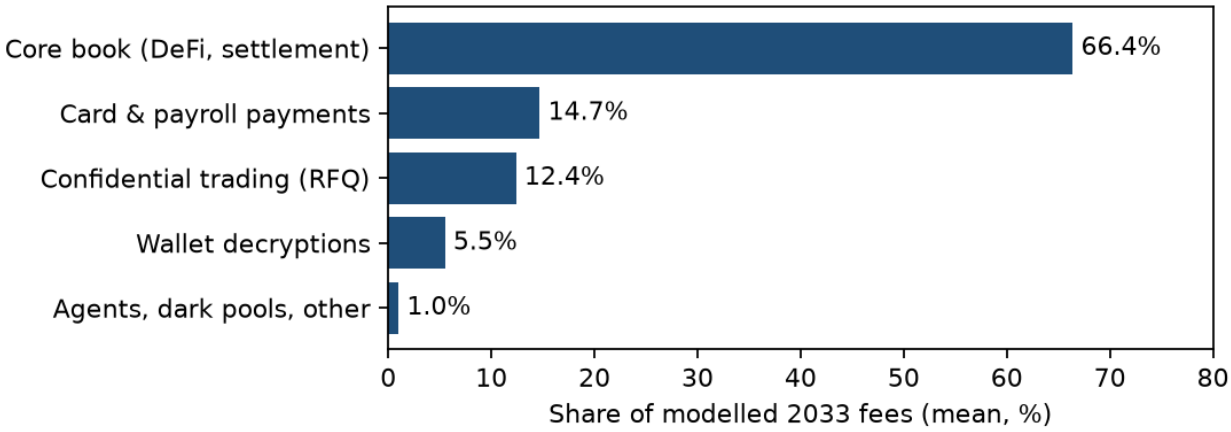
This chapter explains how the headline number is built. It matters because every what-if later in the report is measured against it.

The model simulates 30,000 futures in each of three crypto "worlds" and blends them by weight. In each future it draws the size of each fee stream in 2033, how many tokens exist, how many are locked in staking and as collateral, and how much of the fee flow is burned, then solves for the price at which buyers and sellers of ZAMA balance.



The single most important switch inside the model is whether confidentiality becomes the **default** for tokenised assets and stablecoins on public chains, or stays something users opt into. In the "on" state, 60–95% of that value is confidential; in the "off" state, 2–10%. The chance of "on" is 60% in the trend world, 80% in the super-cycle world and 92% in the adoption world. Those odds rest on Zama's stated goal; the T-REX and Apex news now enters only through the share of tokenised value that settles through Zama (chapter 4).

Where the 2033 fees come from in the model (mean shares):



World	Weight	Middle price (P50)	Median 2033 fees a year
Trend	50%	$749	$72B
Super-cycle	31.25%	$1,977	$176B
Adoption	18.75%	$2,842	$216B
Blend (headline)	100%	**$1,333**	

The most heavily weighted world, trend, comes out well below the blended headline at $749. The blend is higher because the two stronger worlds pull it up.

Median modelled 2033 fees per year: about $72B in the trend world, $176B in the super-cycle world and $216B in the adoption world. The year in which fees alone cover the network's security costs, letting new token issuance step down, comes out around 2030. The model uses only fees in Zama's litepaper: encrypted-

input checks, decryptions, per-transfer fees and a basis-point fee on confidential trades. It charges nothing for compute, licensing or hosting.

Outside reference points for each link estimates, context only

None of these feed the model. They show where each link sits against what others forecast or report. All were fetched on 27 Oct 2026.

Link	Model (P10 / P50 / P90)	Outside reference	Reading
1. Tokenised + stablecoins, 2033	$21.3T / $28.3T / $43.4T	BCG with Ripple: $18.9T tokenised by 2033 (range $12.5–23.4T). Standard Chartered with Synpulse: $30.1T by 2034. Citi: $5.5T in 2030 ($2.7–8.2T). McKinsey: about $2T in 2030 ($1–4T). Stablecoins: Citi $1.9T base and $4.0T bull for 2030	P50 is above BCG's base and close to Standard Chartered. P10 is near the top of BCG's range. The model is on the optimistic side of published forecasts, which fits a success-conditional model
2. Confidential share	4.3% / 54% / 58%	No published forecast. T-REX Ledger: confidentiality is "an opt-in feature … not enabled by default". Circle's Arc plans opt-in privacy on a TEE (testnet stage)	Today's evidence points to opt-in, which is the P10 world. P50 needs the default to flip. This is the least anchored link
3. Fees per year	$39B / $123B / $247B	Visa FY2025: $40.0B net revenue on $16.7T payments volume (about 24bp), 257.5B transactions. Ethereum 2025 fees: about $549M (secondary source, low weight). Chainlink 2025 revenue: about $24.6M (DefiLlama)	P10 is about one Visa (Visa's FY2025 net revenue). P50 is about 3× Visa's revenue today and about 220× all of Ethereum's 2025 fees. The bp figures differ in base: Visa's is per dollar moved, the model's is per dollar secured per year
4. Locked supply	51–52% of 13.2–13.4B	Staked today: 2,350M, about 66% of the 3.55B circulating. Of that, about 692M (29%) sits in operator vaults (KMS 517M, coprocessor 175M) inference	Different bases: today's 66% is of circulating supply; the model's 52% is of all supply in 2033, after the unlocks. Since 2.4, operator stake is bonded in the model apart from yield-driven stake

6. Supply	13.2–13.4B	On chain today: 11.34B total, burn waiting about 88,000 ZAMA	Issuance outruns burn until fees pay for security. Burn removes under 1% by 2033 at the 2.4 price, so supply ends about 17% above today

Upside beyond finance **illustrative, not in the headline**

Each line multiplies an assumed yearly volume by the litepaper's dollar prices ($0.01 per encrypted input, $0.001 per decryption at the deepest discount). The volumes are round assumptions for scale, not forecasts. None of this feeds the $1,333.

Case	Why FHE fits (and the rival)	Illustrative fee line
AI agents paying and negotiating on chain	Agents need to act on balances and bids they cannot reveal to each other. TEEs are the rival, and they are cheaper today	1B actions/day × $0.01 ≈ $3.7B/yr
Identity and credentials (age, KYC, residency)	Checks against encrypted attributes held on a shared ledger. ZK proofs already do the one-party version well	5B checks/yr × $0.01 ≈ $50M/yr
Voting (public and corporate)	Tallying encrypted ballots on public infrastructure. Volumes are tiny, and trust in election rules matters more than cost	1B ballots/yr × $0.01 ≈ $10M/yr
Healthcare data queries	Researchers compute on records that never decrypt. Regulation is slow, and TEEs and federated learning compete	50B queries/yr × $0.01 ≈ $0.5B/yr
Zama's KMS (threshold decryption) sold outside crypto	A 13-party key service with no single holder, useful wherever a cloud KMS is a single point of trust. Cloud KMS from the big providers is the incumbent	1T decryptions/yr × $0.001 ≈ $1B/yr

Both sides. The total, about $5B a year, is around 4% of the P50 fee line. So the case for ZAMA does not rest on these markets, and they do not rescue a failed finance case. They could matter in a different way: each one is an independent route for FHE hardware and tooling to scale, which lowers per-op costs for every lane. Against that, in most of these jobs a TEE or ZK solution is good enough and already shipping. We would need to see a named non-crypto customer paying for Zama's KMS before counting any of it.

4. Two standards, one stack: ERC-7984 and T-REX

This chapter covers how confidentiality is meant to reach regulated assets through the standards they already use. It matters because it is the most direct route from a technology to trillions of dollars of tokenised securities.

The plan is simple: the leading standard for compliant securities tokens (ERC-3643, T-REX) adopts Zama's FHE as its privacy layer, so every issuer on it can switch confidentiality on without rebuilding. The dated record:

Date	Source	What it says
24 Mar 2026	Zama blog	T-REX Ledger builders inherit confidentiality "as a default, not as an upgrade"
24 Mar 2026	T-REX Network release	"privacy as a core infrastructure ... rather than a standalone feature"
~Jul 2026	OpenZeppelin CTO, EthCC	"integrating Zama into 3643"
12 Oct 2026	T-REX Network on X	Ledger mainnet "targeting November"
27 Oct 2026	T-REX docs and FAQ	"Privacy is opt-in, not the default"; the privacy layer activates in the 2026–2035 "Scale" phase

The newest and most detailed source says opt-in. That corrects our earlier wording, which leaned on "default".

> **Thesis: opt-in is a stage.** T-REX chose FHE, and specifically Zama, as its privacy layer, not a TEE, MPC or Canton. Apex, a fund administrator, has committed $100B of assets to the T-REX Ledger by June 2027 stated plan. Regulated systems almost always launch new features as opt-in and make them default once proven. HTTPS took about two decades to go from optional to default on the web.
>
> **Antithesis.** Opt-in may stay opt-in. Asset managers can keep cap tables private off-chain and never pay for on-chain encryption. Other privacy vendors (Inco, Partisia, Dusk, SODA Labs) are members of the ERC-3643 association too. A $100B book of funds that rarely trades produces few fee events.
>
> **Synthesis.** The choice of technology has been made; the timing of activation has not. We read opt-in as timing.

> **Decision recorded (version 2.3).** The small upward adjustment to the default odds made earlier on 27 Oct has been reversed. T-REX's own documents say privacy is opt-in, so the T-REX choice of FHE has no mechanism that acts on whether confidentiality becomes the default. **Overlap: fixed via the evidence ledger.** Each signal now feeds one input, through its mechanism. T-REX and Apex now sit only in the Gateway share, where a standard choosing a privacy rail directly sets how much tokenised value routes through Zama. What-if (d) no longer adds to it. We dropped a separate "delay" test, because the model's fee ramp already contains the delay and testing it again would count it twice.

> **How evidence enters the model.** Every signal is first asked: what technical mechanism does it act on? It then feeds exactly one input, the one that mechanism moves. Signals that build on each other form a chain inside that input: a standard picks Zama, then integrations ship, then live value moves, then fees are paid. Each later link confirms the earlier one, and none is added as a separate bump. News without a code, contract, fee path or standard behind it, however encouraging, is context only and does not move a number. The full list is kept in an evidence ledger beside the model.

Standards-layer depth. We now track a category separate from partnerships: being part of the standard itself. Zama co-wrote ERC-7984 with OpenZeppelin, which also maintains the confidential-contracts library used by the Zerion wallet and the vaults. On 26 Oct 2026 S&P Global completed its acquisition of OpenZeppelin, which now sits inside S&P Global Ratings verified: S&P press release, 26 Oct 2026. The maintainer of the confidential token standard is now owned by one of the firms that rates institutional finance. This is indirect: S&P has said nothing about FHE or confidential tokens. But embedding at the standards level is harder to displace than any single integration. Signposts: S&P or OpenZeppelin naming confidential tokens in ratings or benchmarks; new OpenZeppelin audits of Zama.

5. Who is building it: network, board and hires

This chapter reads the plan written in people. It matters because hires and board seats usually show strategy months before announcements do.

- **Board.** Kyle Samani (Multicoin) wrote on 5 Feb 2026 that he is "staying on the Zama board", and on 2 Oct that his seat bars him from buying ZAMA on the open market verified. His 2024 writing is the clearest statement of the technical thesis: "for simple payments, ZKPs have proven sufficient. However, for any transaction involving shared state—which includes by definition all DeFi—ZKPs are somewhere between insufficient and extremely difficult to implement." (Multicoin, 7 Mar 2024) Counterweight: Multicoin also invested in Fhenix, a rival FHE stack.
- **Hiring sequence.** Cryptographers first, then enterprise sales, then product (a chief product officer from Fireblocks), and in 2026 operators from exchanges, stablecoins and tokenised assets: a real-world-asset (RWA) and decentralised-finance (DeFi) lead from Kraken (tokenised stocks, Solana-heavy) and a technical lead who ran wallets at Tether and Kraken, already committing code. A new chief business officer joined in October.
- **Acquisitions.** Kakarot (Nov 2025), whose co-founder now leads the Solana port, and TokenOps (May 2026), a token-distribution and vesting platform.
- **Size.** About 120 people in January 2026 (self-stated); around 130–170 now inference.
- **Gap.** A head of regulatory and compliance was advertised in January 2026; the posting is gone and no holder is named. We found no ex-regulator hire.

6. Partnerships by depth

This chapter grades partners by how deep the link is, not by logo. It matters because a logo and a live integration handling money are very different evidence.

Depth	Partners
Live, handling value	Morpho vaults and curators; Dfns; Bron (payroll since 16 Jan 2026); TokenOps (owned); Agora AUSD; Merkl (rewards paid as confidential ZAMA since 19 Oct); KAIO (Laser Digital/Nomura-backed, confidential fund distributions since 14 May); **Zerion wallet (27 Oct)**; BitGo (confidential tokens supported)
Integration built	Utila, CMTAT (23 Jul), Blockscout, Yield.xyz, Raycash (testnet), Orion (building), T-REX (in progress)
Service attached	Elliptic (28 Jul), Hypernative, the 13 key-management operators, Conduit, Figment
Listing or logo	Revolut, Kraken listing, Uphold listing, historic logos
Standards layer	ERC-7984 with OpenZeppelin (now part of S&P Global Ratings); ERC-3643/T-REX privacy layer

Zerion. Zerion announced support on its own blog on 17 Sep 2026; Zama announced it on 27 Oct 2026. The Zerion wallet shows confidential token balances on Ethereum natively. Users decrypt with a signed, expiring permit that never gives Zerion lasting access. Confidential vault deposits in the wallet are coming "in the weeks ahead", according to Zerion's CEO Evgeny Yurtaev in Zama's 27 Oct post stated plan. It is the first mainstream self-custody wallet to support confidential tokens.

Partners we did not find: no Circle, Chainalysis or TRM partnership. MetaMask, Rabby, Phantom and Ledger Live do not yet support confidential tokens natively.

7. Real value today: tokenised assets and confidential vaults

This chapter measures what is actually confidential on chain today. It matters because it is the one place where the thesis meets real money now.

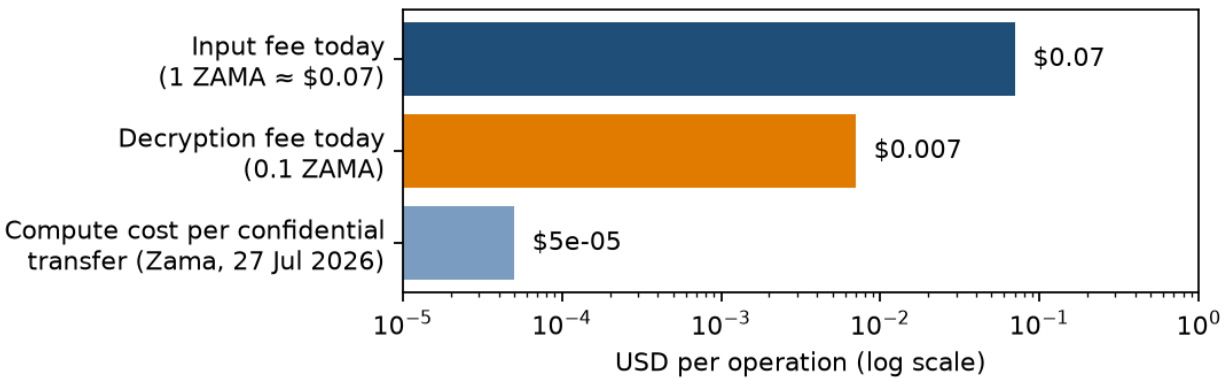
- 16 confidential Morpho vaults since 15 Sep 2026; Morpho cited "$100M+" on 15 Oct; an independent research firm (Alea Research) put Zama's confidential value at $120.2M on 17 Oct verified: Morpho and Alea posts on X.
- Counter-data: an independent privacy tracker found, to 28 Sep, $336.9M total vault assets but only **$35.1M actually confidential** (as of 28 Sep; not rechecked in this edition), with 96% of deposits traceable at the moment of wrapping. A design fix (an asynchronous wrapper with decoys) is published but not shipped.
- Value is concentrated (about 86% with a handful of depositors), partly incentive-driven, and includes Zama's own $5M treasury.

The incentive cliff. Merkl incentives end on 3 Nov (one vault), 8 Nov (most of the rest) and 12 Nov (the Flowdesk High Yield USDT campaign, per Merkl). Those vaults hold about 61% of confidential value. Compare confidential value on 15 Nov with ~$120M on 17 Oct. Reading in advance: if money returns to ordinary transparent vaults, that is timing (pull at today's maturity is low). If it reappears on a rival privacy rail, that counts against.

8. The fee meter, its price and the burn

This chapter shows how Zama actually charges today, and why the fee is a price Zama chooses, not a cost it must recover. It matters because fees are the only source of value for the token in this model.

- On chain today: **1 ZAMA per encrypted input checked, 0.1 ZAMA per decryption**, set 16 Dec 2025 verified. At today's price that is about $0.07 and $0.007.
- The litepaper instead describes prices fixed in US dollars through a price oracle, with plan discounts down to $0.001 per decryption. The two do not yet match. The model uses the litepaper's dollar prices; chapter 13 tests fees fixed in ZAMA.
- Fees flow to a contract with no owner that can only send to the burn address verified. Zama's CEO, Rand Hindi: "100% of all fees … to the protocol" (29 Jul 2026).
- **Only one burn has ever happened:** 17 Jul 2026, 09:01 UTC, about 91,000 ZAMA (≈$6.6k at the price on the day; ≈$6.4k at today's $0.0706), a test. Routine burning has not started; about 88,000 ZAMA (87,934 on 27 Oct) is waiting in the collector. Hindi, 22 Sep: "What matters at this stage is growth, not burn." This is a wording correction only; the model's burn timing changes the price by under 1%.



Zama wrote on 27 Jul 2026 that the compute cost of a confidential transfer is under $0.00005. Today's on-chain input fee (about $0.07) is about 1,400 times that. Against the litepaper's dollar prices, which the model uses, the gap is smaller: about 200 times at $0.01 per input and about 20 times at the $0.001 decryption floor. The figure probably leaves out operator and key-management overhead, but the gap is large enough to mean that revenue depends on pricing choices and volume, not on cost.

> **Thesis: grow first, charge later.** With a price-over-cost gap of roughly 20–1,400 times, depending on which fee, Zama can afford to give away decryptions and cut prices to win the market, as Hindi says: "This year our focus is integrations. Then we will focus on growing revenue" (16 Jul 2026). Infrastructure that becomes the default can charge later; the web's own history (free browsers, paid infrastructure) supports it.
>
> **Antithesis: charge now.** Habits set early are hard to reverse. A protocol that trains its users to expect free decryptions may never bill them, and the token's value rests only on fees. Low prices also mean little to burn for years.
>
> **Synthesis.** The input meter stays paid on every path; what is open is decryption. The model already carries a spread of billing paths, from fully billed to never billed. We treat "grow first" as the base, and watch for a pricing announcement.

Encrypted inputs without proofs: the ZKPoK (zero-knowledge proof of knowledge) removal

What this shows and why it matters: every confidential transaction starts with an **encrypted input**, for example the amount in a private transfer. How that input gets into the system decides how fast and cheap Zama is for wallets and phones, how quantum-safe it is, and what the input fee is charged on. Zama is redesigning this step now.

How it works today. When a user encrypts a value, their app also produces a **ZKPoK** (zero-knowledge proof of knowledge): a short mathematical proof that "I know what is inside this encrypted value, and it is well formed", without revealing the value. The proof stops anyone from copying someone else's encrypted value off the public chain and reusing it (a **replay**), or tweaking it into a related value (**malleability**), and then getting the network to compute on or decrypt data they never knew. The user's device generates the proof. Zama's coprocessors (the servers that run the encrypted computation) check it on the Gateway (Zama's coordination chain), and the input fee, 1 ZAMA per input today, is charged at that moment verified: InputVerification.sol, verifyProofRequest, line 203; litepaper fee model. The litepaper itself names the weak spot: "the ZKPoK is not [post-quantum] (similar to most ZK-SNARKs)" verified: Zama litepaper. It is the one piece of Zama's own cryptography that a future quantum computer could break.

What replaces it. On 9 Oct 2026 Zama co-founder Rand Hindi wrote: "We are getting rid of zkpoks entirely from the protocol. Fewer components = better security", and that Zama is "moving to an OTP protocol for inputs" stated plan. OTP means **one-time pad**: the oldest provably unbreakable cipher, where a message is combined with a random

secret of the same length that is used exactly once. The new flow, already in Zama's public code verified: draft code, fhevm #4620, #4578, #4558; tfhe-rs #3859, #4098:

1. The coprocessors generate a random pad while it stays encrypted, so none of them sees it.
2. The key-management nodes (the 13 operators who jointly hold the decryption key) unlock the pad only for the requesting user, sent encrypted to that user's own key. The user's app needs matching pieces from 9 of the 13 nodes to rebuild it.
3. The app mixes the value with the pad (a simple bitwise XOR) and posts the result, tied to the sender, the target app and a one-time label.
4. The coprocessors mix the posted value with their encrypted copy of the pad. That turns it into an ordinary encrypted value the network can compute on. This conversion is called **transciphering**.

No proof is needed, because there is nothing left to prove. Any posted value unmasks to some valid number, so it cannot be malformed. Only the user who asked for the pad could have produced it, and the one-time label plus the sender binding are checked on chain, which blocks copying and replay.

> **Thesis: fewer parts, safer and lighter.** Removing a complex proof system removes a whole class of bugs. It also removes the one non-quantum-safe link in Zama's own input path. Phones, wallets such as Zerion, and AI agents no longer have to do heavy proving on the device.
>
> **Antithesis, steelmanned.** A proof is checked by mathematics alone. The pad is only secret as long as fewer than 5 of the 13 key nodes collude, and every input now needs the network online before the user can even encrypt: one more round trip, one more thing that can stall. A coprocessor can also withhold pads, which Zama's own code lists as an accepted risk. Nor is it "fully post-quantum": the signatures that authorise requests are classical, as are Ethereum's own.
>
> **Synthesis.** The group that could peek at a pad can already decrypt everything on the network, so secrecy rests on the same trust as before, not on a new one. What changes is liveness (the network has to answer) rather than secrecy. On quantum safety, input confidentiality becomes quantum-safe once pads travel over post-quantum encryption, which is a separate change that is still open (kms #922). Input authorisation stays on classical signatures, like everything else on Ethereum. The old proof only mattered at the moment of proving; it could never leak past data. So the gain is real but narrower than "now fully post-quantum".

What it means for the fee. Today the fee is charged when the proof is checked. In the new code path there is no proof check, and so far **no fee hook at all** verified: no payment code in the pad-path pull requests. Our reading inference is that the fee moves rather than disappears. Each pad costs the key nodes a joint decryption, which is real work per

input, and the natural place to charge is the pad request. The new pad service already counts a budget per requester, which is the obvious hook. Three things are still open: whether a pad is priced like an input (1 ZAMA) or like a decryption (0.1 ZAMA); whether it is charged per value or per package (one package can hold up to 255 values); and whether the pad path ships before the fee does. Cheaper, lighter inputs could also raise volume. The model is unchanged: it already charges per input and carries a range of prices.

Other effects. Workload: coprocessors stop checking proofs and instead generate pads and transcipher, which are light encrypted operations. The key nodes take on one more decryption per input, and Zama is already making that step cheaper (kms #962) verified: code. Solana: Zama's code keeps a Solana input-proof path only "until transciphering exists" verified: fhevm #4408, 13 Oct 2026. The new input format carries no chain ID, so one input design can serve every chain, which simplifies ports beyond Ethereum inference.

Where it stands (27 Oct 2026). The core cryptography is merged and released (tfhe-rs 1.8.0, 17 Sep 2026). The protocol plumbing is open and marked experimental: the pad service still returns mock output, the relayer route is a stub, and the old proof path stays the default. It targets the next protocol version, v0.16. Code review has already caught and fixed a threshold bug, in which one compromised node could have chosen a user's pad verified. Zama has given no ship date unknown.

Signposts: fhevm #4620, #4578 and #4558 merged; a real relayer pad route; a price for pad requests in the Gateway payment contract; a pull request that removes `verifyProofRequest`; v0.16 release notes; kms #922 (post-quantum transport) merged; Solana inputs moving off proofs. Unknown: device proving time today, pad round-trip latency, the billing unit, and the ship date.

9. Decryption in a plain-web world

This chapter covers the single largest open question for value capture. It matters because decryptions are a large share of paid operations.

- Decryptions were about 71% of paid operations historically, and about 49% over the last 7 days verified.
- On Zama's Solana development branch, a design decision marked "adopted" (merged 15 Oct 2026) sends decryptions over ordinary web requests (HTTP) to the key operators, and says plainly that "no Gateway fee applies".
- On the main code line, the same web service also accepts Ethereum requests, with no fee logic. Access is gated by API keys and signatures. The default developer kit still sends decryptions through the paid Gateway.
- Zama has said nothing about how decryption will be priced.

Our reading: this is a value-capture question, not an adoption one. It may be scaffolding to get a new chain working end to end, priced later, or a move toward "free reads, paid writes", like many chains. Inputs stay metered either way. The model already gives 40% weight to decryptions staying billed through the Gateway, 38% to

off-Gateway credits with 25% leakage, 7% to never billed, and 15% to new pricing forms. What-if (a) in chapter 13 moves 10 more points away from the Gateway. Signposts: fee code in the web service, a pricing proposal, the developer kit switching its default path, end of support for the old relayer SDK (software development kit) on 6 Nov.

10. Solana

This chapter covers Zama's second big chain. It matters because Solana carries much of crypto's trading and launch activity, and is also where the rivals are strongest.

- About 300 code changes titled "solana" in Zama's main repository plus 25 in the key-management repository, mostly by Kakarot's co-founder. Prototype February–April 2026; end-to-end June; preview environment September; development-network transactions 7 Oct. The work is on a branch, not yet merged verified.
- Hindi: "End of year for GPU, Solana soon after" (18 Aug) and "Coming in 27" (15 Sep). **Base timing: early 2027** stated plan. This supersedes the litepaper's "H2 2026".
- On Solana, encrypted inputs pay the Gateway fee; decryptions currently do not.
- Rivals there: Solana's own ZK confidential balances (switched off for about a year after a proof bug, re-enabled June 2026, transfers only); Arcium (MPC), live since June.

Through the two-question test, Solana is the one place where a rival is partly winning the same job today. So we treat Solana's timing as early 2027 and expect a smaller Zama share there than on Ethereum, with fees from inputs only. Zama's edge on Solana is shared-state computation (order books, lending, sealed bids), not meme-coin transfers. What-if (c) tests it.

11. Second-order effects: following the dominoes

This chapter is deliberately exploratory. It follows moves that could compound beyond their direct fees, and thinks through how each could cascade. None of it is in the headline.

TokenOps as a distribution funnel. Start with the most mechanical one. TokenOps, which Zama bought in May and relaunched on 7 Oct as "confidential by default", runs token vesting, airdrops and rewards for issuers, with over $2B and 500+ distributions handled historically. Each confidential vest or airdrop hands ERC-7984 tokens to employees, investors and community members who never chose Zama. They now hold something only a confidential-aware wallet can show. That creates pull for wallets, which creates pull for apps where confidential tokens can be used: vaults, confidential trading, payroll. The fees from a vest are tiny. The value is in the users it creates without marketing. The weak link: issuers can still choose ordinary tokens at claim

time, and Zama subsidises free claims. And it is Ethereum-only today; the funnel only matters at scale once it reaches chains like Solana, where launch volume lives.

The wallet cascade. Zerion built on an open standard and Zama's developer kit, which offers "confidential token support" to "any wallet, exchange, or dApp that integrates it" (7 May). Once one wallet has shown the pattern (display, expiring permits, no custody change), the next one's cost is small. BitGo already supports confidential tokens. The real question is whether demand forces MetaMask, Phantom or Ledger to follow, which brings us back to the funnel: the more people hold confidential tokens, the more wallets must show them. Funnel and cascade feed each other. Today it is one wallet, balances only.

Rewards that stay confidential. Merkl now pays vault rewards as confidential ZAMA. Incentive spending turns into confidential holders who can stake or redeposit without leaving the system. Small, and the incentives end within days.

Ratings meeting confidentiality. If S&P's on-chain risk work ever treats "confidential with auditor access" as a control rather than a risk, confidential share classes become rating-positive for tokenised funds. Low odds, very large effect if it happens, and slow.

Native confidential stablecoins. The biggest prize is a stablecoin issuer minting confidential tokens directly instead of wrapping. Zama has public code work toward Circle compatibility, and a published wrapper design names native issuer minting as the clean path. This is also the one place where Circle's own TEE route could win the job outright. It is the sharpest test of the end-state thesis.

The standard without Zama. The dark mirror of standards-layer depth: iExec runs ERC-7984 on TEE. An open standard spreads faster, but it can spread with someone else earning the fees. The model already assumes 10–17% of confidential volume leaks to other stacks; this is the mechanism.

Confidential unlocks. Zama wrote in May that it "will distribute" team and investor tokens through TokenOps, confidentially. If it does, the market would see dates (the cliff is 2 Feb 2027) but not amounts. Chapter 14 explains why this has not happened yet.

Solana as the template for any chain. The Solana port forces Zama to make its system chain-agnostic, which makes the next chain cheaper. Code tests on Polygon (17 Sep) suggest the spread continues.

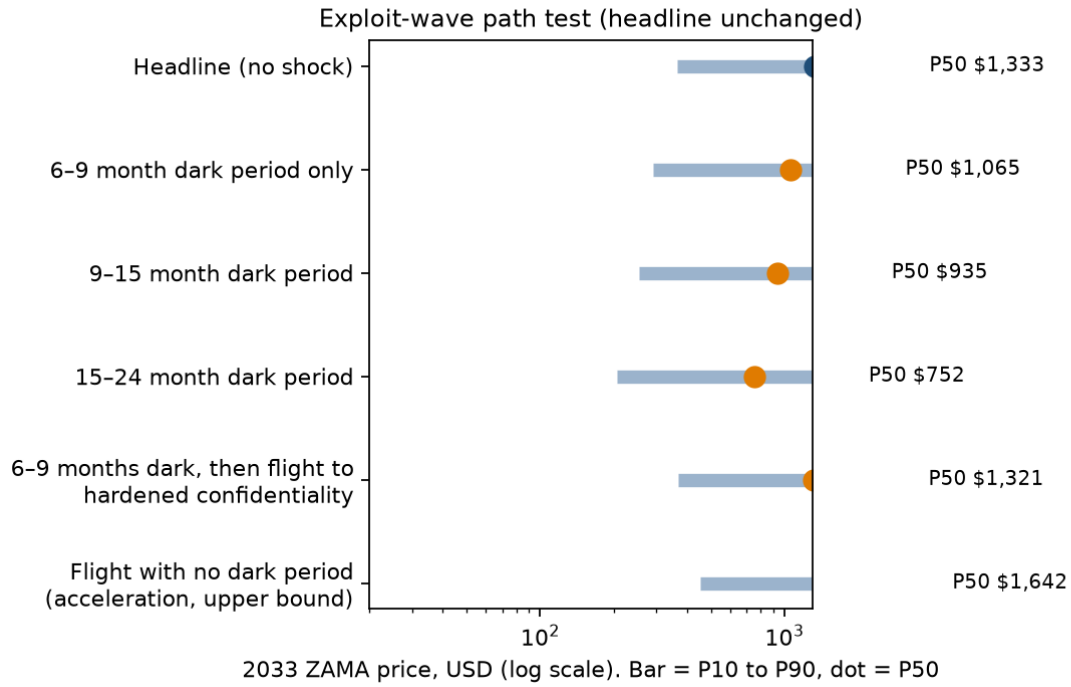
12. Fear or reasoning: a 6–9 month wave of AI-driven exploits

This chapter tests what a crisis would do to the path. It matters because a shock does not only delay adoption; it can change what people adopt.

The scenario: AI systems that find software bugs at scale trigger a 6–9 month wave of exploits (Zama's CEO wrote on 22 Oct 2026: "Brace yourselves, the next 6-9 months are going to be extremely volatile!", adding that the result will be "much, much more secure systems"). The question is where trust goes after the panic.

Think it through. An AI bug-finder attacks the weakest trust points first: hardware enclaves (TEE.fail showed the cost is under $1,000), operator keys held in one place, and cryptography that will not survive quantum computers. Lattice-based FHE has a different shape. Data stays encrypted even while being computed on, so there is no plaintext to pull out of a chip, and the data encryption is lattice-based and expected to be post-quantum (the per-input proof and the signatures are not yet; chapter 8). The remaining attack surface is the threshold key, spread across 13 operators.

> **Thesis: flight to reason.** After a wave of breaches, institutions and regulators demand confidentiality that survives a compromised machine and a future quantum computer. The stack with no secret to steal from a chip becomes the standard. The crisis accelerates FHE.
>
> **Antithesis: flight to the familiar.** Fear does not reason. Under stress, people run to what they know: custodians, closed systems, banks' private ledgers, hardware they already bought. Fewer moving parts feels safer than new cryptography. A crisis also starves new technology of capital, and a token can lose 70–95% before the dust settles. Historically, panics have entrenched incumbents as often as they have displaced them.
>
> **Synthesis: when fear is the likely outcome, let reasoning lead rather than instinct.** Both flights will happen, in sequence. The first flight is to the familiar; the lasting one is to what actually survived. The question is not whether people are afraid, but whether reasoning leads once the panic peaks. The signposts below are how we will know which way capital moves.



Path (same 2033 success world, rerun on the current model, 30,000 draws per world)	**P10**	**P50**	**P90**	**Chance ≥ $2k**	**vs headline**
Headline, no shock	$363	$1,333	$3,776	32.5%	—
6–9 month dark period, no change in what is adopted	$291	$1,065	$3,012	23.4%	−20.1%
9–15 month dark period	$255	$935	$2,644	18.5%	−29.9%
15–24 month dark period	$206	$752	$2,131	11.7%	−43.6%
6–9 months dark, then flight to hardened confidentiality	$368	**$1,321**	$3,744	32.3%	−1.0%
Acceleration: flight with no dark period (upper bound)	$452	**$1,642**	$4,691	41.7%	+23.1%

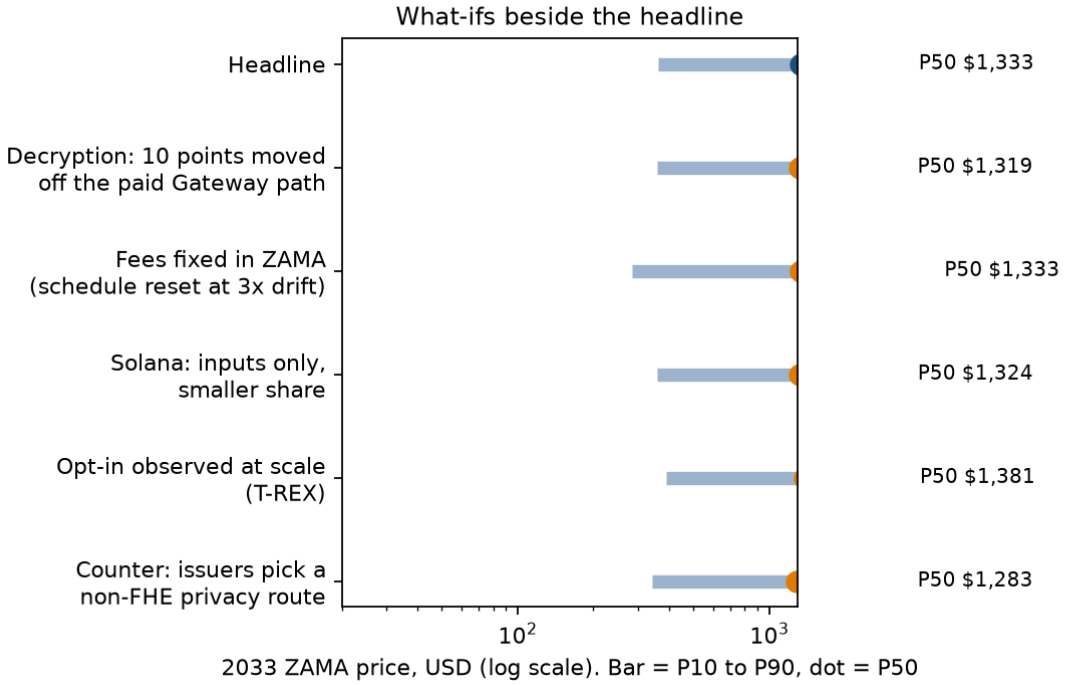
Method: a delay of D years places 2033 where the undelayed path stood D years earlier, with late-stage fee growth of 25–45% a year. "Flight" adds 3–10 points to the share of public tokenised value that settles through Zama, raises the core fee book by 10–50%, and raises the chance that a major card issuer settles confidentially by 8 points. These are path tests inside the success world. They never enter the headline. The flight effects are our assumptions, not measured results, so the size of the recovery below follows from what we assumed.

The result is the point: a dark period alone costs 20–44%, and, if our assumed flight effects hold, a flight to hardened confidentiality buys almost all of it back. The fear outcome is costly; the reasoned outcome nets out close to the headline, and an early flight is clearly above it. Note that these figures replace earlier ones from an older version of the model ($2,155 and $2,698; version 2.3: $1,253 and $1,562), which should no longer be quoted.

Signposts: exploit post-mortems that blame hardware enclaves or key custody; institutions or regulators citing post-quantum confidentiality afterwards; where capital goes once the panic peaks.

13. What-ifs beside the headline

This chapter tests six questions with model runs, plus confidential perps, argued without a run, that are open but not settled enough to put into the headline. Each is an argument; the numbers show the size, not a forecast. None changes the headline.



What-if (30,000 draws per world)	P10	P50	P90	≥ $2k	vs headline
(a) Decryption: 10 points moved from paid Gateway to off-Gateway credits	$358	$1,319	$3,732	32.0%	−1.1%
(a) stress version: 20 points moved	$354	$1,303	$3,681	31.5%	−2.3%
(b) Fees fixed in ZAMA, schedule reset when it drifts 3x	$284	$1,333	$4,774	36.0%	0% (wider)
(b) same, reset only at 10x drift	$247	$1,333	$5,034	36.0%	0% (wider)
(c) Solana: inputs only, smaller share	$360	$1,324	$3,741	32.1%	−0.7%
(d) Opt-in observed at scale on T-REX	$390	$1,381	$3,817	33.4%	+3.6%
(d) counter: issuers pick a non-FHE privacy route	$341	$1,283	$3,735	31.5%	−3.8%
(a′) Holders net-add 5–15% of the free float to stake each year (free stake only)	$370	$1,351	$3,774	33.1%	+1.4%
(e) Fees on parked value, high case $26B a year	$379	$1,514	$4,254	38.2%	+13.5%

(a) Decryption priced off the Gateway

Claim: the new web path takes more decryptions off the paid route. **Chain:** developer kit switches to the web path → fewer decryptions billed at full price → a smaller fee flow. **Evidence:** chapter 9. **Result:** small, −1.1% (−2.3% for twice the move), because the model already spreads decryption billing over four paths and inputs stay paid. **Counter:** the large price-over-cost gap (20–1,400x) means free decryption can be a deliberate growth choice, with billing added later through credits or plans. **Signposts:** fee code in the web service; a pricing proposal.

(b) Fees fixed in ZAMA instead of in dollars

Claim: if fees stay set in ZAMA rather than pegged to dollars, the fee in dollars moves with the token price. **Chain:** higher price → higher dollar fee → some users pay more, some leave (we assume demand elasticity 0.55–1.0) → fee flow and price move together. Zama's board resets the schedule when it drifts too far. **Evidence:** the on-chain schedule is in ZAMA today; the litepaper describes dollar pricing. **Result:** the middle is unchanged by construction (the schedule is set at the middle price), but the range widens: P10 falls to $247–284 and P90 rises to $4,774–5,034. Fixed-token fees make the token more reflexive in both directions. **Counter:** the board controls the schedule and has every reason to move to dollar pricing as the litepaper describes. **Signpost:** activation of a price oracle in the fee contracts.

(c) Solana: inputs only, smaller share

Claim: Zama earns only from inputs on Solana, and Arcium and native ZK keep part of the market. **How tested:** the model has no separate Solana lane, so we approximated: 5 points more weight on decryptions never being billed (from the Gateway path), and 20% lower Zama shares of AI-agent payments and dark-pool trading. **Result:** −0.7%. Solana matters for the narrative and for users more than for the 2033 fee arithmetic. **Counter:** Solana could become a large source of new users, especially with a funnel like TokenOps. **Signposts:** merge to main, an audit report, a public testnet, Solana's Breakpoint conference 15–17 Nov.

(d) Opt-in observed at scale

Claim: issuers on the T-REX standard actually switch privacy on at scale, so confidentiality starts to look like the default for tokenised assets, the way websites drifted to HTTPS. **How tested:** this is the one place the default odds may move on T-REX, and only on observed behaviour: raise them by a small step (+0.2 on the log-odds scale). Nothing else changes; the T-REX choice of FHE is already counted once, in the Gateway share. **Result:** +3.6%. **Counter:** issuers pick a non-FHE privacy route (Canton share classes, a TEE or ZK option), the same step down: −3.8%. **Signposts:** T-REX Ledger mainnet in November; Zama contracts deployed on it; the share of new issues with privacy switched on.

(a′) Holders lock more of the float into stake

Claim: outside holders keep moving free tokens into stake, 5–15% of the free float a year, which shrinks the float that can sell. **How tested:** mapped onto the existing stake floor, on free stake only, over 3.1 years, so operator stake (change c) is counted once. **Evidence** on chain, attribution inferred **:** over the last 28 days stake rose 178.9M, but 128.9M of that was one Foundation-style batch and 9.3M went to Zama-named vaults; outside holders added at most 40.7M, and only 8.8M in the last 14 days. From June to October they added about 77M, at most about 15% a year of the float, and lumpily (September was −105M). Staked ZAMA has almost no liquid market, and unbonding takes 7 days (the manager role can raise this to 365). **Result:** +1.4%. **Counter:** the flow is small, lumpy and partly reward-driven; it could reverse when campaigns end. **Signpost:** outside-holder stake growth after the November campaign ends.

(e) high case: fees on parked value at $26B a year

Claim: curator revenue shares and stablecoin yield kickbacks reach $26B a year at P50 scale. **Result:** +13.5%, still well below the flow meter. **Counter:** no rate, contract or proposal exists, and confidential Morpho vault fees are zero today (chapter "How to read this report"). **Signpost:** a confidential vault fee recipient pointing to a Zama or burn address; an issuer announcing a revenue share.

> ### (e) Confidential perps what-if, not in the headline
>
> **Claim.** Perpetual futures ("perps") are now where crypto's speculative demand lives. Transparent perp books leak positions and liquidation prices. The end state of on-chain perps is hidden positions, which means computing on everyone's encrypted shared state. That is FHE's job, and it would create a new fee stream for the Zama Protocol.
>
> **Causal chain.** Traders move to on-chain perps → visible positions get hunted, copied or traded against → large traders demand privacy → venues add confidential positions → margin, funding and liquidation checks must still run across all traders' hidden data → FHE (publicly verifiable and composable) → fees for each encrypted input and decryption → ZAMA burned.
>
> **Evidence.** On-chain perp volume runs at about $620–700B a month, about half the October 2025 peak but roughly twice the same months of 2025 secondary: CoinGecko, The Block, DefiLlama . Exchanges without a central operator (decentralised exchanges) went from about 3% to about 10% of all perp volume between early 2025 and April 2026. Memecoin launch revenue fell about 80% while stock and commodity perps grew. In 2025 a trader's public 40× position and its liquidation price were visible to everyone; he alleged he was hunted (alleged, not proven). Binance's founder called in June 2025 for a "dark pool perp DEX" (a decentralised exchange for perpetual futures). Zama's own products today are spot only: Confidential RFQ and Confidential Swap, live since 15 Sep 2026, with all swap fees used to buy back and

burn ZAMA verified: Zama blog. Confidential perps exist only as test-network builds in Zama's ecosystem, including a winner of Zama's developer bounty verified: code. We found no first-party Zama perp product or roadmap item.

Strongest counter-argument. Hyperliquid, the leader, argues its transparency is a feature: open books stop the operator from exploiting order flow. Perps are dominated by high-frequency trading, and FHE is roughly two to three orders of magnitude short of matching-engine speed today. MPC dark pools (Renegade, Arcium) are live or close to it, and operator-trust "hidden orders" (Aster) are cheaper. The best-known manipulation case (JELLY, 2025) was an oracle trick that privacy would not have stopped. Perp volume is also in a cyclical decline.

Synthesis. Speed is timing. A hybrid design, with a public price, hidden positions, encrypted liquidation checks and entry through RFQ, fits today's throughput. No live MPC or TEE system offers composable margin and liquidation on shared hidden state, so they do not yet win the same job. Transparency's defenders have a real point, so the likely end state is a confidential layer for size beside public books, not a replacement. This is a credible upside path but not a stated Zama plan, so it stays a what-if.

Illustrative fee scenarios only: our assumptions, not a model run, not in the headline

Scenario	Confidential perp volume a day	Encrypted inputs a day (orders, edits and cancels)	ZAMA a year	At $0.07
A. Niche	$50M	25k	≈9.7M	≈$0.7M
B. 1% of today's on-chain perp volume	$210M	210k	≈81M	≈$5.7M
C. 10% of on-chain perps (end-state-like)	$2.1B	2.1M	≈810M	≈$57M

Assumptions: average order $5–10k; five fee events per order (edits and cancels); fee held at today's 1 ZAMA per input plus three decryptions per fill at 0.1 ZAMA. The board can reprice fees in dollars, and batching or pad pricing (chapter 8) could cut per-order fees sharply. Scenario C would equal about 7% of supply a year if all of it were burned.

Signposts. Zama or a top-10 perp exchange announcing confidential positions on Zama; Confidential RFQ adding perps or tokenised-stock pairs; a Zama-ecosystem perp reaching mainnet with more than $10M in open positions; Hyperliquid or Lighter proposing privacy, or choosing MPC or TEE instead (against).

14. Risks, the token unlock, and a correction

This chapter collects what could go wrong for the token specifically. It matters because a good technology does not guarantee a good token.

Unlock = path and timing, not end-state value. The headline ($1,333 in 2033) comes from fees paid in a mature market. Whether insider tokens arrive in one lump, monthly or daily changes how the price may move over the next few months. It does not change the 2033 fee base, so the headline does not depend on it. The cadence detail below is a near-term price-path matter.

The hard source: Zama's litepaper. It states: "20% of the supply will be circulating at TGE [token generation event, the launch], with the rest subject to lockups as follows" verified: litepaper, read 27 Oct 2026 :

Group	Share	Lock-up	Cliff	Unlocked at cliff
Treasury	20%	2 years	None	–
Growth	10%	4 years	None	–
Team	20%	4 years	1 year	1/10th
VCs	20%	2 years	1 year	1/10th
Angels	10%	2 years	1 year	1/10th

With launch on 2 Feb 2026 and 11,000M genesis supply, this fixes **550M unlocking on 2 Feb 2027** (one tenth of team, VCs and angels). CoinMarketCap and the independent dashboard agree with it. Two readings contradict it and carry lower weight: Tokenomics.com (longer insider lock-ups) and DefiLlama's token rating (no cliff lump at all, below). The only open point is the release cadence after the cliff.

> **Clarification: the February 2027 unlock.** We earlier suggested that team and investor tokens vesting confidentially through TokenOps could soften the cliff on 2 Feb 2027. That overstated it. Confidential distribution would hide amounts; it would not reduce how many tokens unlock. Whether Zama uses it for insider tokens is unproven either way. Its May post says it "will distribute" through TokenOps (future tense), and nothing is due to move before the cliff. The original minting wallet still holds about 7.67B ZAMA, roughly 68% of supply, and has not moved since 27 Jan 2026 verified via block explorer. That is expected, because nothing in it has unlocked, so its silence is not evidence. Our figure is **about 914M ZAMA unlocking in February 2027: a 550M cliff on 2 Feb (about 565M that day including one day of linear release)**, or about 665M if releases come in monthly events. Signposts: transfers out of the minting wallet on or after the cliff, and where they go.

- **Concentrated custody.** One wallet holding two-thirds of supply is a single point of failure until distribution happens.
- **Lock-ups are promises, not code.** The token contract has no lock or vesting logic, so the lock-ups are enforced only by the Foundation honouring them. Tokens could in principle move early, or be held back late. Holders rely on Zama's word and on public on-chain monitoring.
- **Threshold-key risk.** Confidentiality rests on no more than 4 of the 13 key operators colluding; 9 of 13 must take part to decrypt verified: Zama KMS docs. One global key means a key compromise would expose history, not just one account inference. Zama's CEO notes FHE is used "only for privacy, so breaking FHE would not allow stealing funds" (9 Oct).
- **The proof on each input is not yet post-quantum;** its removal is planned.
- **Thin organic usage.** After removing Zama's own test traffic, organic Gateway fees run at about $6.6k a year, with a ceiling of about $11.9k (27 Oct; the model's live band since 2.4). The model's 2033 middle cases are $72–216B a year, about ten to thirty million times more. That means fees must grow roughly 15× every year for six years. The case for it is the switch from opt-in to default: if tokenised assets and stablecoins become confidential by default, the fee base is the value moving on public chains, not today's test traffic. If that switch does not happen, this gap is the strongest single argument against the headline.
- **Incentive cliff** on 3, 8 and 12 Nov (chapter 7).
- **Token unlock:** ≈914M ZAMA becomes transferable in February 2027 (a 550M cliff on 2 Feb, about 565M that day including one day of linear release), roughly a quarter of today's ~3.55B circulating supply by the dashboard's count, then ≈411–425M a month to February 2028. Unlocked does not mean sold; the size of any selling cannot be derived from the sources.
- **Licensed stacks and rivals** using the same standard without paying Zama.

The vesting schedule, rebuilt from primary sources

Launch (TGE) was 2 Feb 2026 verified: Zama launch post. Genesis supply 11,000M. Shares, lock-up lengths, cliffs and the one-tenth cliff release come from the litepaper's lock-up table verified: litepaper. Two things the litepaper does not state, so we infer them: that each lock-up runs from TGE and includes the cliff, and that releases after the cliff are spread evenly by day. Both match the published schedule used by the independent dashboard.

Category	Share	ZAMA (M)	Lock-up, cliff	Schedule	Feb 2027 (M)
Public sale, TGE campaigns, liquidity	20%	2,200	None	Fully unlocked at TGE	0
Treasury	20%	2,200	2 years, no cliff	Linear, 2 Feb 2026 to 2 Feb 2028 (≈3.01M a day)	≈84
Growth	10%	1,100	4 years, no cliff	Linear, 2 Feb 2026 to 2 Feb 2030 (≈0.75M a day)	≈21
Team	20%	2,200	4 years, 1-year cliff, 1/10 at cliff	220M on 2 Feb 2027, then 1,980M linear to 2 Feb 2030 (≈1.81M a day)	220 + ≈47
VCs	20%	2,200	2 years, 1-year cliff, 1/10 at cliff	220M on 2 Feb 2027, then 1,980M linear to 2 Feb 2028 (≈5.42M a day)	220 + ≈141
Angels	10%	1,100	2 years, 1-year cliff, 1/10 at cliff	110M on 2 Feb 2027, then 990M linear to 2 Feb 2028 (≈2.71M a day)	110 + ≈71
February 2027 total					**≈914**

Our figure, by part. (1) **550M** from the team, venture-capital (VC) and angel cliff on 2 Feb 2027: the amount is verified from the litepaper (one tenth of 5,500M); the date is consistent across the litepaper, CoinMarketCap (which states it builds schedules with project teams) and the dashboard, and treated as firm (TGE plus one year). (2) **≈259M** of insider linear releases over the rest of February (3–28 Feb, 26 days): inferred, because the daily-linear shape is not stated in the litepaper. (3) **≈105M** from the treasury and growth streams for the whole month (28 days), which have been running since launch: inferred on the same basis. **550M + 259M + 105M ≈ 914M in February 2027.** From March 2027 the same streams give about 411M (30-day months) to 425M (31-day months) a month until the VC, angel and treasury schedules end on 2 Feb 2028; after that, about 77M a month (team and growth) until 2030.

How the secondary figures reconcile. These are not competing estimates; they count different things on the same schedule. The independent dashboard's forecast of 914.0M for February counts everything above for the calendar month, so it equals our figure. Alea Research's 809M counts only the team, VC and angel allocations (cliff plus their February linear releases: 550M + ≈259M ≈ 809M) and leaves out treasury and growth. A figure of 550M counts only the cliff day itself. Nothing in the sources lets us

derive how much will actually be sold: unlocked is not sold, and the minting wallet still holds the insider tokens, so the mechanics of release (for example through TokenOps) are not yet visible on chain.

A day-level cross-check (CoinMarketCap). CoinMarketCap's unlock page shows 565.21M ZAMA for 2 Feb 2027 alone: VCs 225.42M, Team 221.80M, Angels 112.71M, Treasury 3.01M, Staking rewards 1.50M and Growth 0.75M. Every line matches our rebuild to the hundredth of a million. Each insider line is its one-tenth cliff plus one day of linear release: VCs 220 + 5.42, Team 220 + 1.81, Angels 110 + 2.71. Treasury and growth are single days of their linear streams. CoinMarketCap labels the VC line "linear", but its size is a cliff plus one day, not a catch-up. The cliff date (2 Feb 2027), the start at TGE and the cliff sizes are consistent across the litepaper, CoinMarketCap (which states it builds schedules with project teams) and the dashboard; we treat them as firm. The release cadence after the cliff, daily or monthly, stays open: the only releases observed so far, the Foundation's batches, were monthly. The "Staking rewards" line is new issuance at about 1.5M a day (5% a year on 11B). It does not come from locked allocations and is not part of the 914M.

Do rewards on locked tokens stack on top of the unlock?

Thesis. Locked team and investor tokens might be staked and earning rewards, so that more than the schedule unlocks in February. **Finding: contradicted on chain.**

- **Mechanics (verified, Zama docs).** Staking means approving and depositing ZAMA into an operator's staking contract, which moves the tokens. Delegators receive a liquid share token, can claim rewards at any time, and wait 7 days to unbond. Rewards come from issuance, initially 5% a year, split 60% to KMS nodes and 40% to coprocessors, with an operator commission of at most 20%.
- **Custody (verified, on chain).** The token contract (ZamaERC20) has no lock, vesting or escrow functions, only mint, burn, pause-minting and role controls. The minting wallet received 9,680M at deployment on 13 Nov 2025. It sent 2,015M out before TGE, the last on 27 Jan 2026, and has sent nothing since. It holds 7,665M today. Nothing in it has unlocked yet, so that silence carries no signal. That wallet is not a vesting contract, so the locked allocations sit there unstaked, earn no rewards, and are released by transfer rather than by code.
- **Docs on rewards for locked allocations (unknown).** The litepaper and docs say nothing about rewards on locked allocations. Since nothing locked is staked, the question does not arise today.
- **Effect on the gap (none).** 550M, 809M and 914M differ only in scope (see above). Rewards on locked tokens add 0M. Separately, staking issuance adds about 42M of new supply in February. It goes to current stakers and is not an unlock.

Elements still inferred, after a second, wider search. Each element is marked verified, contradicted or unknown, with the sources checked.

- **Lock-up lengths (verified, primary).** The litepaper's lock-up chart shows VCs and angels fully vested at year 2 and the team at year 4, which matches its table. Tokenomics.com ("data verified by the Zama team", last updated May 2026) instead lists VCs and angels at a 12-month cliff plus 24 months linear and the team at 12 + 48 months, ending 2 Feb 2031. contradicted by the primary source The litepaper chart and table, CoinMarketCap and the independent dashboard all agree on the shorter lock-ups, so we keep them.
- **Lock-up start at TGE, cliff date and cliff sizes (treated as firm).** Consistent across the litepaper, CoinMarketCap (which states it builds schedules with project teams) and the dashboard; CoinMarketCap matches to the day. Daily vs monthly cadence remains open (only monthly releases observed so far).
- **Release shape: daily or monthly (unknown for insiders by design until 2 Feb 2027).** Sources split. Daily: CoinMarketCap, the independent dashboard and CoinLaunch (about 3.77M a day for treasury and growth). Tokenomist calls treasury vesting "linear", and its unlocked count (about 2,553M) fits continuous accrual from search snippets; page blocked. Monthly: Tokenomics.com (61 events on the 2nd; 2 Feb 2027 ≈ 665M). CryptoRank hides its frequency behind a login. DefiLlama has no ZAMA schedule, and Messari's page gave no content. The team, VC and angel tokens have not unlocked yet; their cliff is 2 Feb 2027. Their release shape therefore cannot be observed before then, and the minting wallet's silence is not evidence either way. The only behaviour we can observe is the Foundation's. Batches of about 119M went out from a Safe multisig on 22 Jun, 3 Jul, 5 Aug, 7 Sep and 7 Oct 2026, each split across 16–18 operator staking vaults on chain; the sender being the Foundation is inferred. This matches Zama's 2 Mar 2026 post: "All unlocked Treasury and Growth tokens from February have been staked." So treasury and growth tokens move in roughly monthly batches, on no fixed day, and are staked. The batch size does not quite fit the schedule. About 119M a month is above the 105M (28 days) to 117M (31 days) that the linear streams produce, and the February batch should be about 105M. Possible reasons are rounding in our reading, rewards included in the batch, or a different start date unresolved. This is one custodian's operational choice about its own tokens. That is an analogy for how insider tokens might be released, not evidence about them analogy only. **Open until the cliff:** ≈914M (daily schedule, calendar month) versus ≈665M (monthly event basis). The headline does not change.
- **DefiLlama's reading (lower weight; contradicts the litepaper).** DefiLlama's token rating flags an "undisclosed unlock" and models no lump: treasury over 22 months, growth over 46, VCs and angels a 12-month cliff then 12 months linear, team 36 months, giving about 336M a month from February 2027, roughly 13% of today's float each month secondary: search snippets only. It has no source for dropping the one-tenth cliff, which the litepaper states, so we keep it as a lower-weight interpretation. On total size it lands close to our schedule; it differs on shape.

DefiLlama has published no unlock study of its own; the large study built on its data is BullpenBrief's (box below).

- **MiCA (the EU crypto-asset regulation) white paper (verified; silent on vesting).** It is listed in the register of ESMA (the EU securities regulator) (Central Bank of Ireland; updated version notified 14 Jan 2026). It says the company controls about 30% of the initial supply (treasury 20% plus growth 10%) and that the token has no transfer restrictions. It covers custody risk only in general terms and gives no cliff, vesting or release shape.
- **Sale terms (verified).** The public auction sold 880M with no lock-up; CoinList's terms state 100% unlocked at TGE. The litepaper's 12% public-sale share (1,320M) equals the 11,000M genesis supply less the 9,680M minted to the minting wallet, which suggests the remaining 1,320M was minted elsewhere for the sale and TGE inferred.
- **What the minting wallet holds (inferred).** It holds 7,665M, while team, VCs and angels total 5,500M. The roughly 2,165M surplus is most likely unvested treasury and growth tokens inferred. The Foundation's custody accounts were pre-funded with about 1,203M before May, but treasury and growth releases reach about 1,370M by February 2027. So the minting wallet should send tokens to Foundation custody before the cliff, probably around late 2026, even though no insider tokens unlock. That transfer is a testable signpost. It would not be an insider unlock.
- **Custodian and tooling (partly verified).** Bron smart accounts (EIP-7702) hold large stakes: the largest holds about 417M, 17.6% of all stake (likely treasury, inferred). The monthly staking batches, by contrast, come from a Safe multisig, not from Bron accounts. On 27 Jan 2026 a Bron custody account wrapped 1,000M ZAMA into the confidential wrapper contract; amounts held inside the confidential token are encrypted, so public trackers cannot see these balances. Rand Hindi (7 Oct 2026): "We acquired TokenOps a few months back to enable confidential vesting and distribution." Zama dates the acquisition to May. Neither names TokenOps as the tool for Zama's own insider releases unknown.
- **"Nothing released" (contradicted).** Earlier text said no release had happened. The minting wallet holds the insider stack, which has not unlocked yet, so it is expected to be still. It is false for treasury and growth, which unlock monthly from Foundation custody and are staked (about 800M total stake by 2 Mar, about 34% of circulating supply). Part of today's stake is therefore Foundation stake, not outside delegators. Rewards accrue on that unlocked stake. They do not accrue on the locked insider allocations, which remain unstaked.
- **Final vesting date (verified: 2030, not 2031).** The MiCA white paper (E.12) says "the full Token supply will become tradable within 4 years". CryptoRank's countdown ends 2 Feb 2030. Both match the litepaper. Tokenomics.com's 2031 end date is contradicted.

Searched: Zama docs, litepaper and MiCA white paper (E.12, G.05); Zama blog (2 Mar 2026 token update); Rand Hindi and Zama posts on X about unlock, vesting, treasury, custody and TokenOps; CoinMarketCap, Tokenomics.com, Tokenomist (snippets only, page blocked), CryptoRank (schedule behind login), Messari (no content), DefiLlama (token rating, snippets only), CoinLaunch; exchange notices from Binance (Seed Tag), OKX (30 Jan), Bybit (Feb Launchpool), Upbit (14 Apr) and Coinbase, none of which give vesting terms; no full

Binance Research report found; Blockscout transfers for the minting wallet, the Foundation custody accounts, the ConfidentialWrapper and the staking contracts. Dune public queries were not reached; the zero-address mint query timed out on Blockscout.

Sell pressure by holder type

Staking today (verified, 27 Oct 2026). The two protocol staking contracts hold 2,358M ZAMA (KMS 1,526M, coprocessor 832M). That is 21% of the 11,337M total supply (which already includes about 337M of issuance) and about 64% of the 3,672M outside the minting wallet, a figure that includes Foundation custody and Foundation stake. Observed issuance of about 337M in roughly 267 days annualises to about 460M. That gives a gross yield of about 20% on the staked base, or about 16–20% after commission. Both figures are our calculation from on-chain numbers.

Our lean: VCs and the team mostly restake. VCs and team members hold as an infrastructure bet and stake unlocked ZAMA with operators for yield, shrinking the liquid float. Angels are different: we expect some profit-taking (below), which is healthy distribution, not a thesis break.

Evidence for.

- The Series B (June 2025) was led by Pantera and Blockchange. Blockchange called it its "third and largest investment in Zama", and the round was framed as strategic, built on "partnership value rather than capital".
- The Series A was led by Multicoin and Protocol Labs. Multicoin has a long record of staking its liquid holdings in proof-of-stake networks.
- A yield near 20% is high enough to pay a holder to wait. VCs and angels unlock only 10% at the cliff and the rest over 24 months, so a fund that stakes keeps earning while it holds.
- **How fast outside holders add stake (on chain, attribution inferred).** From 16,955 staking events up to 27 Oct 14:17 UTC: over the last 28 days total stake rose 178.9M. Of that, 128.9M was one batch from a Safe multisig (likely the Foundation), 9.3M went to Zama-named operator vaults, and outside holders added at most 40.7M. Over the last 14 days they added 8.8M. From June to October outside holders added about 77M in five months, roughly 15% a year of the float or less, and lumpy: September was −105M. Staked ZAMA (stZAMA) has no DEX or lending venue, so almost none of it is liquid today. Unbonding takes 7 days, and the manager role can change that to up to 365 days. No fee burn has happened yet. This is the evidence behind the absorption what-if in chapter 13. It is too short and lumpy a window to put into the headline.
- **Policy and structure.** Zama has said all unlocked Foundation tokens are staked, which sets the norm. Only one tenth unlocks at the cliff, and the rest is released over 12–36 months while staking pays about 20%, so waiting is paid.

Antithesis (short). Staked is not unsellable: the liquid share token can be sold at once, and unbonding takes 7 days. Funds can stake and short perps at the same time, which is selling in all but name. And angels have large gains (below).

Angels: expected profit-taking, not a thesis break

Sourced token entry prices (verified, primary). The only token sales were the January 2026 public sale (12% of supply, three parts). None of these tokens sit in the 2 Feb 2027 cliff: all were 100% unlocked when claimed on 2 Feb 2026.

Sale	Size, buyers	Price	at $0.05	at $0.07 (today)	at $0.10	at $0.15
Community sale, opened 15 Jan 2026	Up to 220M (2%); max 40,000 per OG NFT, original claiming wallet only (5,500 NFTs given to early creators, developers and testers, none to the team); 3,549 depositors (Zama Dune dashboard)	$0.005	10×	14×	20×	30×
Public auction, 21–24 Jan 2026	880M (8%); 11,103 bidders, $44M paid; clearing price $0.05 (floor $0.005, $55M FDV). Bids below $0.05, including 122M bid at the $0.005 floor, were refunded	$0.05	1.0×	1.4×	2.0×	3.0×
Auction with full bonus	Up to +10% tokens on filled bids (two of: OG NFT, bonus code, Bron wallet)	≈$0.045	1.1×	1.5×	2.2×	3.3×
Pre-TGE sale, 27 Jan–2 Feb 2026	220M (2%); $10k cap per buyer, for unfilled bidders	$0.05	1.0×	1.4×	2.0×	3.0×

Sources: Zama blog, "The Zama OG NFT Claim Portal is Live" (5 Jan 2026) and "Announcing the Zama public auction"; Zama auction results post ($118M committed); docs.zama.org/auction (facts: lock-up none; bonus); The Block (13 Jan 2026); CoinList sale terms (100% unlock at TGE). So yes: some holders entered at $0.005, but they are community (OG NFT) buyers with at most 40,000 tokens each, already fully unlocked and tradable since launch. They are not part of the angel 110M cliff. No Echo/Sonar, Legion, SAFT (simple agreement for future tokens) or OTC (over-the-counter) token round was found.

Assumed insider entry prices (inferred). Zama sold no private token round; its rounds were equity, and the token allocations of early backers are not disclosed. The table converts each round's post-money valuation into a price per token on the 11,000M genesis supply. This assumes token rights in proportion to equity, so treat it as an implied entry price inferred.

Round	Raised, valuation	Implied entry	at $0.05	at $0.07 (today)	at $0.10	at $0.15
Pre-seed, Apr 2020	$1.1M; valuation not disclosed; assumed $7–10M	≈$0.0006–0.0009	55–80×	75–110×	110–155×	165–235×
Seed, Jun 2021	$7.8M; valuation not disclosed; assumed $30–40M	≈$0.0027–0.0036	14–18×	19–26×	28–37×	41–55×
Series A, Mar 2024	$73M at "nearly $400M" (TechCrunch)	≈$0.036	1.4×	1.9×	2.8×	4.1×

Series B, Jun 2025	$57M at over $1B (Decrypt)	≈$0.091	0.55×	0.8×	1.1×	1.6×

Sources: raise amounts from DefiLlama's raises list and Cointime; Series A and B valuations from TechCrunch and Decrypt. Pre-seed and seed valuations are our assumptions inferred. Notable individuals (for example founders of other chains) joined the Series A, so not every "angel" entered at seed prices.

Reading. The earliest angels sit on roughly 20–100× gains, so selling part of the cliff is rational profit-taking. That moves tokens to new holders and broadens distribution. It is a short-term path effect, not a change to the fee thesis. **Size:** the angel cliff is 110M, about $7.7M at $0.07. That is about one day of spot trading (about $8.5M a day across exchanges, DefiLlama) and below the $14–27M of open perp positions. Even if every angel sold on the day, it would be absorbed over days, not weeks, especially if routed through RFQ or OTC. Series A and B investors sit near or below their entry at today's price, which supports the lean that VCs hold and stake rather than sell.

Can the unlock be hedged on perps? (capacity check, inferred) The 2 Feb 2027 cliff is about 550M ZAMA, roughly $38M at $0.07. Total open positions in ZAMA perps across exchanges are only about $14–27M secondary: Swolecharts, perpdexlist; snapshots differ, and Binance holds about 60%. ZAMA is not listed on Hyperliquid. Funding today is slightly positive (about 3–7% a year, paid by longs), so there is no crowded short. Recipients cannot hedge the full cliff on perps without pushing funding sharply negative. Locked tokens cannot be posted as margin, so any hedge needs cash collateral. Zama's Confidential RFQ explicitly names "a VC selling a token allocation" as a target user verified: Zama blog, 23 Jul 2026. An insider exit there would happen off the public order book, and its fees would burn ZAMA. That reduces visible selling but also what we can observe.

Evidence: how unlocks have moved prices. Main weight goes to Keyrock (16,000+ unlocks), Messari, Animoca Research (about 35,000 events) and 6th Man Ventures (about 5,000); BullpenBrief (5,664 cliffs, on DefiLlama data) is lower weight.

- **Pressure comes before the date.** Keyrock finds negative moves in about 90% of cases, starting about 30 days ahead and settling about 14 days after. BullpenBrief finds that for unlocks above 10% of float, most of the drop (−13.7%) comes in the 30 days before, with −6.6% after.
- **Size relative to float matters.** 6th Man: below 1% of float, no measurable effect; above it, negative. Messari: cliffs of 5% of float or more matter most, within about a week of the date. Animoca: a 1% unlock costs about 0.3% in the week before and after; daily linear release has no significant effect. ZAMA's 550M cliff is about 15–20% of float, so it is in the range that matters.
- **Who receives it: studies split.** Keyrock finds team unlocks hurt most (about −25%) and investors more controlled (they hedge or sell OTC); BullpenBrief finds recipient type barely matters.
- **Signpost: January 2027.** If the pattern holds, any pressure shows up from early January in price, negative funding and growing perp shorts. A quiet January is evidence for the restaking lean.

Before SUI's May 2024 unlock, for example, funding turned negative 20 days ahead.

Not found. We found no on-chain evidence yet linking any named VC wallet to staking, and no OTC deals. Insider tokens have not unlocked yet (cliff 2 Feb 2027), so there is nothing to trace until then. Their absence on chain is not a signal.

Signposts.

- The shape of the first insider transfers on and after 2 Feb 2027: one cliff batch followed by monthly batches (Tokenomics.com lists events on the 2nd; supports ≈665M), or daily or continuous streams (supports ≈914M).
- Transfers out of the minting wallet on or after 2 Feb 2027 (a top-up of Foundation custody before then is expected and is not an insider unlock), and whether they go to exchanges, custodians or staking contracts.
- The staked total and the outside-holder staked share in the weeks after 2 Feb 2027. Staking rising by several hundred million from 2,358M, with the outside share holding near or above half, supports the lean; flat or falling staking with exchange inflows refutes it.
- Growth in ZAMA perpetual open interest and negative funding around the cliff, which would signal hedging.
- Sales of staking share tokens on secondary markets.

15. Regulation, and the state as a source of trust

This chapter covers the rules that gate institutional use. It matters because institutions will only adopt confidentiality that regulators accept.

- **US stablecoins:** the GENIUS Act is expected to take effect on 18 Jan 2027.
- **EU anti-money-laundering rules:** whether regulators accept view keys (auditor access to specific encrypted data) as meeting their requirements is the central design question. If rules pushed the odds of default confidentiality down by 10 points, the middle outcome would fall by about 11% (these odds are the second-largest single input in the sensitivity table; the exploit-wave paths in chapter 12 move more, but they change several inputs at once); rules that accepted view keys would make them a moat.
- **Bank capital rules (Basel):** whether confidential tokenised assets can count as the lowest-risk category; small effects either way in earlier tests.
- **Legitimacy:** we judge it likely (about 0.8) that regulators have had substantive contact through an ECB project, ISO/NIST work, a MiCA filing and the Open Tokenization Layer group, but we found no regulator endorsement of view keys. If governments use encrypted-but-auditable systems for benefits or votes, the main effect would be indirect: making confidentiality the trusted default (earlier tests: +1% to +3.5%).

16. Company value versus token value

This chapter separates what Zama the company earns from what the ZAMA token captures. It matters because they are not the same thing.

> **Thesis.** Company and token rise together: the company holds about 30% of supply, and all protocol fees go to adoption and buyback-and-burn by stated policy.
>
> **Antithesis.** Zama's MiCA paper says the token gives no claim on business revenue. Licensing income (Zama has licensed its patents since 2022) is the company's, not the token's. A discretionary company buyback would make the company a central party in the eyes of the US SEC.
>
> **Synthesis.** The token's value is the fee meter. The cleanest way for company deals to reach the token is to route them through the meter (paid in ZAMA via the Gateway), not through buybacks. Earlier tests put these routes at well under 1% of the headline.

17. Signpost calendar

This is what to watch, with dates, and what each would tell us.

When	Watch	What it tells us
3 Nov / 8 Nov / 12 Nov 2026	Vault incentives end; confidential value on 15 Nov (and again after 12 Nov) vs ~$120M on 17 Oct	Pull at today's maturity; against only if value moves to a rival rail
6 Nov 2026	Old relayer SDK end of support	Which decryption path the new default uses
Nov 2026	T-REX Ledger mainnet; Zama contracts; privacy flag	Standard-stack timing
Weeks ahead	Zerion vault deposits ship; the next wallet announces	Wallet cascade
15–17 Nov 2026	Breakpoint (Solana)	Solana timing and rivals
End 2026	GPU coprocessors in production (stated)	Scaling plan on time
Early 2027	Solana launch (stated); audit; merge to main	Second chain
18 Jan 2027	GENIUS Act in effect	US stablecoin rules
Late 2026	Minting wallet tops up Foundation custody for treasury and growth	Custody mechanics (not an insider unlock)
On or after 2 Feb 2027	Insider transfers out of the minting wallet: to exchanges, custodians, staking or confidential vesting	Release shape (settles ≈665M vs ≈914M) and sell pressure (no movement before the cliff is expected and tells us nothing)
2 Feb 2027	Insider cliff 550M (≈565M that day with one day of linear release); ≈914M across February, or ≈665M on a monthly-event basis (date consistent across litepaper, CoinMarketCap and dashboard; treated as firm)	Supply
Any time	Decryption pricing announced; fee code in the web path; price oracle switched on; routine burns start	Value capture
Any time	A named head of regulatory and compliance	Readiness
Any time	Error/comb bootstrapping (ePrint 2026/2630) merged into TFHE-rs; KMS #922 merged	Cheaper ciphertext refresh; post-quantum links between key-system parties. Neither changes the per-op fee meter
Any time	A confidential vault fee recipient set to a Zama or burn address; an issuer revenue share	Verifies (e), fees on parked value
Two quarters after T-REX mainnet	Share of new issues with privacy on	Timing only, unless paired with a non-FHE route winning

Pre-registered "against" signals, written down now so we cannot explain them away later:

- Circle's Arc privacy goes live and carries real USDC confidential volume.
- Arcium usage keeps growing after Zama launches on Solana.
- T-REX adds a non-FHE privacy option, or Apex issuers choose Canton for confidential share classes.
- Fhenix or another FHE stack takes Zama's institutional partners.
- Vault value leaving Zama reappears on a rival privacy rail.
- A rival still wins a job after FHE is fast and cheap.

18. Corrections and transparency

This chapter lists what we got wrong or overstated before, and what replaced it.

Earlier claim	Corrected to
Draft of this edition: version 2.4 headline $187 (P50), from reading "net absorption of 5–30% a year" as float selling	Withdrawn before publication: absorption means holders adding to stake. Turnover stays at 2.3; headline $1,333. The stake reading is what-if (a′), +1.4%
Monthly Foundation staking batches sent on the 2nd of each month through Bron	About 119M each on 22 Jun, 3 Jul, 5 Aug, 7 Sep and 7 Oct, from a multisig, split across 16–18 vaults inferred. Bron accounts hold large standing stakes (largest about 417M, 17.6% of stake, likely treasury, inferred), not the batches
February 2027 release fixed at ≈914M with daily-linear timing; ZKPoK removal a one-line note	≈914M on the primary-source schedule counted by calendar month, ≈665M on Tokenomics.com's monthly-event basis (timing dispute, same total); Tokenomics.com's longer insider lock-ups contradicted by the litepaper chart; rewards on locked tokens contradicted on chain (0M); ZKPoK removal expanded into a full section in chapter 8
T-REX privacy "confidential by default"	Opt-in, not yet live; activation in the 2026–2035 phase (T-REX docs, 27 Oct)
Confidential team vesting could soften the Feb 2027 cliff	Overstated: confidential vesting would hide amounts, not reduce them. Unproven either way until the cliff; the minting wallet's silence is expected and not a signal
Vault value "$40M+"	$100M+ (Morpho, 15 Oct) / $120.2M (Alea Research, 17 Oct); ~$35M truly confidential at 28 Sep (ethprivacy tracker; not rechecked in this review)
Fee $0.01–0.04 per operation	On chain: 1 ZAMA per input (~$0.07), 0.1 ZAMA per decryption; $0.01 is the litepaper plan price
"$1.06 per transaction"	$0.106
Solana "H2 2026", decryption-only branch	~300 changes, preview and development network, not merged; "Coming in 27"

Our 20 Oct post: Solana decryption "free for now, on purpose"	No Zama source says "for now" or "on purpose"; the design is adopted and pricing is unstated
A code branch was securities (ISIN) logic	Generic math functions, stale
Usage quotas were fee groundwork	Rate limiting
Transciphering near-ready	Experimental
A hire was a recent defection to Circle	Left in Oct 2024; joined Circle May 2025
Samani's board seat unknown	Confirmed staying (5 Feb 2026)
A widely shared quote attributed to Samani about FHE	Not found in any source; excluded
Headcount ~150–250	~120 (Jan 2026), ~130–170 now (inference)
Tokenised stocks "mostly on Solana"	Multi-chain, Solana-heavy
Merkl and Shiba Inu on Zama's ecosystem page	Announcements only
CMTAT launch 29 Jul; BitGo a logo	23 Jul; BitGo is a live integration
Zerion support "since 27 Oct 2026"	Zerion's own blog announced it 17 Sep 2026; Zama's announcement and the CEO quote are dated 27 Oct
About 87,000 ZAMA waiting to be burned	87,934 on 27 Oct (about 88,000), read on chain
10,000 and 100,000 transfers/s "on dedicated chips"	Zama's projection assumes FHE gets ~10× faster every 18–24 months; 1,040/s is one GPU node's compute, not end-to-end settlement
Feb 2027 unlock shown as an equal-weight range (550M / 809M / 914M)	Rebuilt from the litepaper: 550M cliff on 2 Feb + ≈259M insider linear + ≈105M treasury and growth ≈ 914M in February. 809M (Alea) is the insider-only part; 550M is the cliff day only (chapter 14)
Arcium "200,000+ a day, not confirmed by Arcium" (v16.1)	Arcium said it in its own 9 Jun 2026 release (1M computations passed, more than 200,000 a day). Its explorer shows 2.70M cumulative on 27 Oct, about 12,000 a day on average since then: a peak, not sustained
"Circle chose a TEE" for Arc (v16.1)	Circle's Arc documentation and privacy white paper plan opt-in privacy on a TEE (AWS Nitro at first). It is a plan at testnet stage, not a live product
Inco "moved to TEE" (v16.1)	Inco now describes itself as a TEE-based private rollup. Its cToken has been live on Base since 2 Sep 2026. That it was earlier an FHE user is our inference

Removing ZKPoKs "planned since 2023 alongside transciphering"	Not sourced; replaced with Hindi's 9 Oct words: "moving to an OTP protocol for inputs"
CEO quoted as "6–9 volatile months"	Exact words (22 Oct 2026): "the next 6-9 months are going to be extremely volatile"
Alea figure "excluding confidential ZAMA"	Not stated in Alea's post; qualifier removed
TEE.fail presented without funding note	The TEE.fail page acknowledges gifts from Qualcomm and Zama; now disclosed
Exploit-wave paths $2,155 / $2,698	Older model; rerun on version 2.4: $1,321 (dark then flight) and $1,642 (acceleration)
Zama's "decentralised" and "verifiable" are true today	Stated plans; today 18 permissioned operators and hardware enclaves in the key system

New signals this edition, each tied to one input

These come from our live on-chain log and public filings (UTC times). No model input was changed. Signals that would need an input change are listed below for approval. `weak` marks thin evidence. News is context only.

Signal	Input it bears on	Reading
Live organic fees ≈$6.6k a year (27 Oct 13:43), up from ≈$2.5k (24 Oct). Ceiling ≈$11.9k	Live organic fee band	Applied in 2.4: band now $6.6k–$11.9k (+0.3% alone). It affects only hand-off timing
Smoke tests are 623 of 868 inputs (72%)	Live organic fee band	Confirmed excluded from the model's live base. Only the $28.8k headline includes them
63 of 270 organic inputs are claims of Zama-funded cZAMA rewards (Merkl)	Live organic fee band	Real actions, but driven by incentives. Read the post-campaign level, not today's
114 of 245 organic inputs hit the cUSDC and cUSDT wrappers or the vault router. Holders: cUSDT 8,830, cUSDC 2,006	Stablecoin confidential share	Early, small, real use of the stablecoin lane
Decryptions are about 71% of paid ops. Solana decryption skips the Gateway, and the plain-web relayer has no fee code yet	Decryption path (share billed)	Shown as a sensitivity: all decryption unbilled moves P50 by −19.4%
iExec runs ERC-7984 on a TEE	Gateway share of confidential value	Applied in 2.4: a 3–10% cut to Zama's share (−0.8% alone)
SEC docket S7-2026-30 (Rule 17ad-7(h)(2), transfer agents)	Confidential-by-default odds	Watch item. No new letters since 26 Oct

OTL and DTCC tokenisation news	none	Context only
Unlock: about 914M on 2 Feb 2027, then 410–425M a month	Price path, not 2033 value	Already in the unlock chapter
Staked about 2,350M (about 66% of 3.55B). About 29% of it is operator vaults inference	Stake	Applied in 2.4: operator stake (5–6.5% of supply) bonded apart from yield-driven stake (+1.2% alone)
A proposal to read net absorption of 5–30% a year as float turnover	Free-float turnover	Withdrawn: absorption means holders adding to stake, not selling. Turnover stays at 2.3. The stake reading at 5–15% is what-if (a′), +1.4%
Zama's CEO: "curator revenue share and stablecoin yield kickbacks" (The Rollup, 16 Sep 2026) stated plan	Core fee line (fees on parked value)	Applied in 2.4 as (e): $2–11.6B a year at P50 scale (+5.2% alone). Verified only when a vault fee recipient or an issuer revenue share shows it
Foundation-style monthly staking batches, about 119M each, on 22 Jun, 3 Jul, 5 Aug, 7 Sep and 7 Oct from a multisig, split across 16–18 vaults inferred	Stake (what-if a′)	Excluded from outside-holder absorption
About 1.0B treasury tokens not accounted for in public trackers weak	Supply	Not used until an on-chain trace confirms it
Perps venues building on confidential order flow	none	Context only, not counted
ZKPoK paper, ePrint 2026/2403 inference	Per-op fee drift	May cut input-proof cost. Not quantified
Confidential RFQ: 15 straight fills, all from a single taker weak	Activity per dollar	Not evidence of a market yet
KMS change #922 (hybrid post-quantum key exchange enforced between key-system parties) still open; DDRop on AWS Nitro untested weak	Security and trust (no input)	Watch items
ePrint 2026/2630: exact error bootstrapping in TFHE, built on TFHE-rs. Refresh 3.5× faster for 8-bit integers, 9.1× with "comb" refresh of several ciphertexts at once verified: abstract	None (watch)	Refresh only: any other function still needs full programmable bootstrapping. Watch for it landing in TFHE-rs. The fee meter does not change, because Gateway fees are per operation. Narrows the FHE-vs-TEE speed gap at the margin
AWS KMS adds ML-KEM key encapsulation (26 Oct 2026) verified: AWS	None (signpost)	Post-quantum key exchange becomes standard in cloud key services. Zama's #922 uses its own TLS library, not AWS KMS, so this is context, not a dependency

ePrint 2026/2634: near-optimal hardness for the closest vector problem even with preprocessing	None (context)	Background support for lattice assumptions
ePrint 2026/2638: decision attacks on "known-covariance" ring-LWE, a variant that also reveals a summary of the secret and errors not verified for TFHE	None (watch)	Breaks a strengthened assumption proposed for threshold encryption, not ordinary ring-LWE. TFHE's GLWE reveals no such hint and uses power-of-two moduli, which the paper says its reduction does not reach. Not shown to apply; unverified until checked against Zama's threshold key system
ePrint 2026/2643: ML-DSA-65 signing key recovered from one power trace on a microcontroller	None (context)	An implementation leak on physical hardware, not a break of the math

Input changes made in version 2.4 (27 Oct 2026): (b) live organic band $6.6k–$11.9k; (c) operator stake separated from free stake; (d) a cut to Zama's share for the TEE version of ERC-7984; (e) fees on parked value, $2–11.6B a year at P50 scale (stated plan). A proposed change (a), reading net absorption as float selling, was withdrawn; its stake-side reading is what-if (a′). Step-by-step effects are in "How to read this report".

Source check for this edition (27 Oct 2026). Re-fetched and verified: Zama's KMS documentation (9 of 13 to decrypt; safe while at most one third collude); S&P Global's 26 Oct 2026 completion release (OpenZeppelin now reports to S&P Global Ratings); Arcium's 9 Jun release and explorer; Inco's Lightning page and 2 Sep blog; Circle Arc privacy docs; BCG/Ripple, Standard Chartered, Citi and McKinsey forecasts; Visa FY2025 results; T-REX Ledger docs and FAQ ("Confidentiality is an opt-in feature … not enabled by default"; mainnet "not yet live"). Not rechecked in this edition: Hindi's posts of 7, 9 and 22 Oct; the MiCA white paper wording (E.12); the ethprivacy $35.1M figure; the burn balance after 27 Oct; Solana timing; and the T-REX "2026–2035 Scale phase" wording.

Glossary

FHE	Fully homomorphic encryption: computing on data while it stays encrypted.
TFHE	The fast FHE scheme Zama uses.
Lattice	The family of math problems FHE and most post-quantum cryptography are built on.
Post-quantum	Expected to stay secure against future quantum computers.
ZK / ZKP	Zero-knowledge proof: proving a statement without revealing the data.
ZKPoK	Zero-knowledge proof of knowledge: the small proof attached to each encrypted input today.
TEE	Trusted execution environment: a sealed, hardware-protected area of a chip.
MPC	Multi-party computation: several computers jointly compute without any one seeing the secret.
Threshold key	A key split among operators. Here 9 of 13 must take part to decrypt, and 5 colluding could reconstruct the key.
KMS	Key management service: Zama's 13 operators holding the threshold decryption key.

Coprocessor	Off-chain computer doing the encrypted math for a host blockchain.
Gateway	Zama's coordination chain where inputs are checked, decryptions requested and fees paid.
Relayer	Service passing app requests to the Gateway or key operators.
Transciphering	Converting conventionally encrypted data into FHE form.
ERC-7984	Open standard for confidential tokens on Ethereum-style chains.
ERC-3643 / T-REX	Standard for compliant tokenised securities; T-REX Network runs the T-REX Ledger.
View key	Permission for a named party (auditor, regulator) to decrypt specific data.
RWA	Real-world assets (funds, bonds, stocks) represented as tokens.
RFQ	Request for quote: a trading venue where prices are requested privately.
Burn	Permanently removing tokens from supply.
Unlock / cliff	Date when locked tokens (team, investors) become transferable.
Perps	Perpetual futures: contracts that track a token's price with leverage and no expiry date.
Funding	The periodic payment between long and short perp holders that keeps the perp price near spot. Negative funding means shorts pay longs, usually a sign of heavy hedging or bearish bets.
Open interest	The total value of open perp positions.
TGE	Token generation event: the token's launch, here 2 Feb 2026.
One-time pad (OTP)	A cipher that mixes a message with a random secret of the same length, used once.
Liquid share token	A token a staker receives for staked ZAMA. It can be traded while the stake stays in place.
Unbonding	The waiting period (7 days) between asking to unstake and getting the tokens back.
Bron	A self-custody wallet and payments company whose smart accounts hold large ZAMA stakes (the largest about 17.6% of all stake, likely treasury; inferred). Also runs confidential payroll on Zama.
MiCA	The EU's Markets in Crypto-Assets regulation; token issuers file a white paper under it.
Canton	A permissioned ledger network used by banks, private by sharing data only with trade parties.
Basis point	One hundredth of one percent (0.01%).
EIP-7702	An Ethereum upgrade that lets an ordinary wallet act as a programmable smart account.
World	One of the model's three crypto scenarios (trend, super-cycle, adoption), blended by weight.

P10 / P50 / P90	Outcomes at the 10th, 50th (middle) and 90th percentile of simulated futures.
What-if	A tested scenario shown beside the headline, never inside it.
Stated plan	Something Zama or a partner says it will do and is visibly building.

Sources include Zama's litepaper, blog (246 posts read and indexed) and public code; T-REX Network docs and releases; S&P Global press release (26 Oct 2026); tee.fail and Intel and AMD security bulletins (28 Oct 2025); Gentry (STOC 2009); Regev (STOC 2005); NIST FIPS 203 (2024); BIS Bulletin No. 58 and BIS working papers; Kinexys publications; on-chain data from Ethereum and the Gateway; independent trackers (Alea, ethprivacy.tools, Dune). Model: a seeded Monte Carlo of 30,000 draws per world; version 2.4 (27 Oct 2026), with each piece of evidence counted once (chapter 4). Not financial advice.